Thermoelectrically Cooled HgCdTe Reference Device

Engineering Assessment

Revision 3

1 November 2026

Hypothetical reference article. Numerical release incomplete.

Abstract. A source-referenced nine-pin, two-ear housing and an independently selected three-stage cooler envelope define a synthetic HgCdTe detector article. This assessment combines two distinct numerical descriptions. A genuine coupled three-dimensional solid calculation resolves 88 thermoelectric legs, internal conductors, a header, explicit joint and attach layers, and the detector. At 0.1 A, 300 K underside bath and 2 mW imposed die heat, its principal fine mesh gives approximately 265.6 K die temperature and 0.552 V. Current and conservative heat checks pass, while original raw boundary-gradient requirements remain unmet: the principal ceramic/attach mismatch is 6.48% against 5%, and the best independent normal-refinement bath error is 1.069% against 1%. Later transverse cases remain near 1.29%. No passing combined mesh and systematically refined comparator have established numerical release.

A separate five-temperature model uses bounded nonlinear one-dimensional P/N transport, corrected wire routes, a configurable sink, imposed detector heating and preliminary gas/radiation exchange. Its enclosure is an explicitly filled stepped-core proxy inside candidate cap/window solids, not a fully resolved native cavity. At nominal 300 K, 0.15 A, 30 kPa krypton and 2 mW die heat, this model predicts approximately 232.2 K and 1.152 V. Its 52 accepted and 11 rejected states remain a separate comparison. A further preregistered material screening admits 141 of 181 reduced cases. With fixed 300 K hot ceramic, 2 mW die heat and wires ON, but enclosure and nonlocal leakage omitted, the nominal analog admits 186.484 K at 0.4 A and 0.970800 W input. This is a sampled conditional result, not demonstrated assembly feasibility. A boundary-matched manufacturer comparison shows substantial lift and voltage disagreement. Above 300 K material continuation is assumed; no measured device-accuracy interval is established.

Engineering interpretation. Source geometry, physical hypotheses, numerical verification and empirical accuracy are treated separately. Numerical digits identify conditional calculations, not manufacturing tolerances. Neither the VIGO housing plus candidate cooler reference nor the reduced enclosure model is a verified commercial assembly. The report supports bounded engineering exploration; it does not certify a device or a numerical release.

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1 Reference article, model hierarchy and interpretation

The reference article is a three-stage thermoelectric cooler supporting a small HgCdTe die inside a nine-pin TO-66-style package. The intended questions concern detector temperature, electrical demand, heat rejection, parasitic paths and sensitivity to uncertain inputs. A proprietary assembly is not reconstructed, and agreement with a manufacturer's limiting numbers is not an acceptance requirement for this synthetic article.

Description	Modeled physics and boundaries	Important exclusions
Three-dimensional solids	Coupled temperature and potential; 88 legs; 216 solid domains; 0.1 A; 300 K Kovar underside; 2 mW die source	External Cu/Au links OFF; no cap, gas, radiation, optics or detector carriers
Reduced nonlinear package	Five unknown temperatures; nonlinear P/N profiles; Cu/Au conduction; Cu Joule heat; effective sink; detector heat	Ideal stage contacts and simplified bypass; no spatial spreading or die gradients
Candidate enclosure solids	Native cap/roof/window BREP and standalone 8,143-tetrahedron fixture	Geometry only; no conformal connection to the 88-leg mesh and no new thermal field
Preliminary exchange proxy	Closed cylinder and filled stepped core; gray radiosity; pressure-dependent gas chord network	Actual interleg voids, pin/wire occlusion, full native-surface view factors, gas diffusion/convection and spectral transmission absent

1.1 Input evidence and physical hypotheses

Input family	Evidence category	Interpretation
Housing envelope/pin functions	Published drawing	Source-backed external dimensions; some contour details and seals inferred [1]
Cooler envelope	Independent datasheet	Candidate dimensions, not proof of assembly compatibility [2]
Stage split and routing	Inferred construction	28/12/4 couples and internal paths are explicit hypotheses
P/N transport	Research analog	Measured specimen tables to 300 K; assumed extension to 390 K [4]
Joints, contacts, cap/window	Engineering assumptions	Explicit dimensions; unknown grades and contact impedances
Gas, pressure, emissivity, loads, sink	Operating scenarios	Selectable conditions, not observations or statistical priors

1.2 Meaning of verification

A converged algebraic solve, current conservation, an accurate controlled benchmark and empirical device validation answer different questions. The first three can support a numerical implementation without identifying real materials or package construction. Raw-gradient qualification is stricter than a conservative reaction balance. Chosen low/nominal/high scenarios are engineering alternatives, not confidence intervals. The sections below keep these distinctions visible for every result.

2 Geometry and source interpretation

The governing package reference is the VIGO family drawing ZTM-TO66-Z030, Revision 9, dated 1 March 2024 [1]. Its two mounting ears and nine C pins of nominal diameter 0.76 mm govern the corrected header mesh. Overall flange dimensions are 31.5 by 17.5 mm, mounting-hole pitch 24.4 mm and nominal hole diameter 3.7 mm. The source does not uniquely establish every fillet, seal or internal dimension. Inferred contour radii and idealized glass seals remain explicit modeling assumptions.

The TEC Microsystems 3MC04-044-05 datasheet [2] independently provides a candidate three-stage cooler envelope: hot side 6.4 mm, cold side 2.4 mm and height 3.8 mm. **The VIGO housing plus this cooler reference is not an identified OEM assembly.** Internal artwork, stage populations and material grades are unverified.

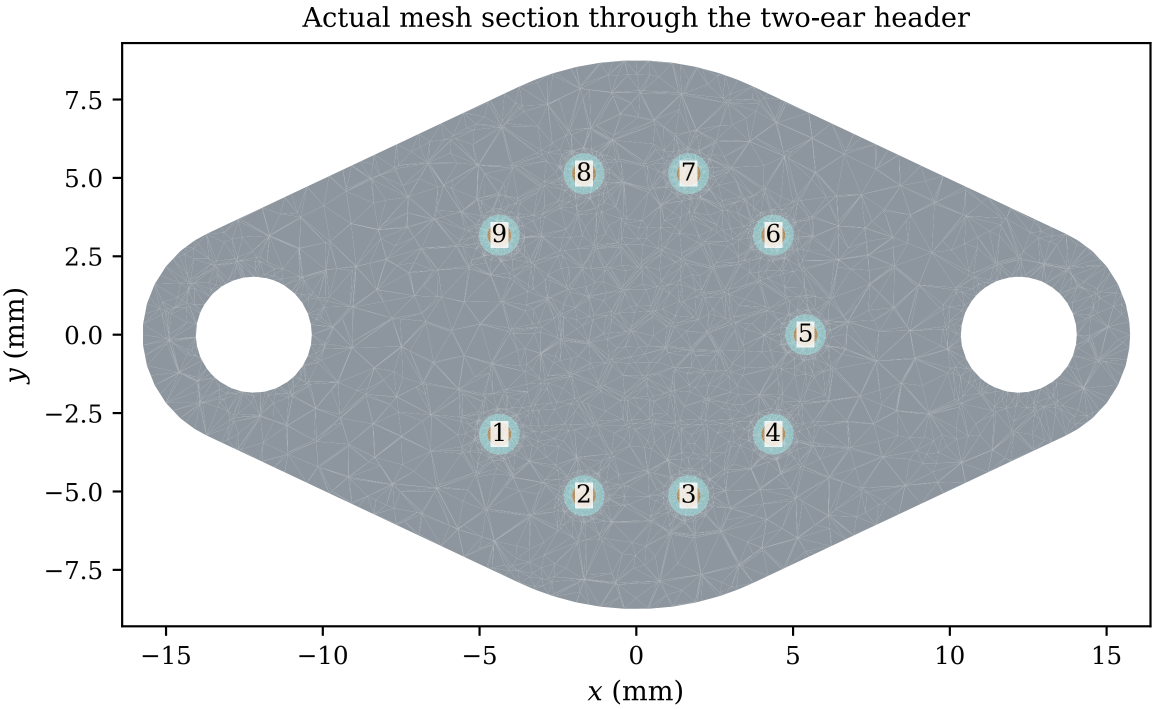

Figure 1: Actual corrected tetrahedral mesh cut at header mid-height. Labels denote pin numbers in the modeled top-view coordinate system; the source bottom view is mirrored consistently. White mounting holes and the two-ear outline are mesh geometry, not an ellipse or an artist's reconstruction.

Element	Provenance and interpretation
Header and C pins	Drawing envelope; Kovar and glass thermal properties are proxies.
Pin functions	1/9 TEC supply/return; 7/8 detector; 5/6 thermistor positions.
Internal TEC	28/12/4 couples, totaling 44 pairs and 88 legs; assumed routing.
Detector and attach	1 mm^2 by 200 μm die; 30 μm attach; hypotheses.
Hot joint	Explicit 50 μm metallic layer; unidentified alloy.

3 Actual solid topology and discretization

The corrected mesh contains 216 solid domains. All 88 legs form one electrical series route through the copper pads and interstage conductors. There are two module terminals on the *hot-side* copper pads. The external pin-to-pad copper and detector gold links are disabled in the present calculation: the terminal drive acts directly on the resolved module pad faces. The nine pins remain passive thermal solids with exposed ends adiabatic.

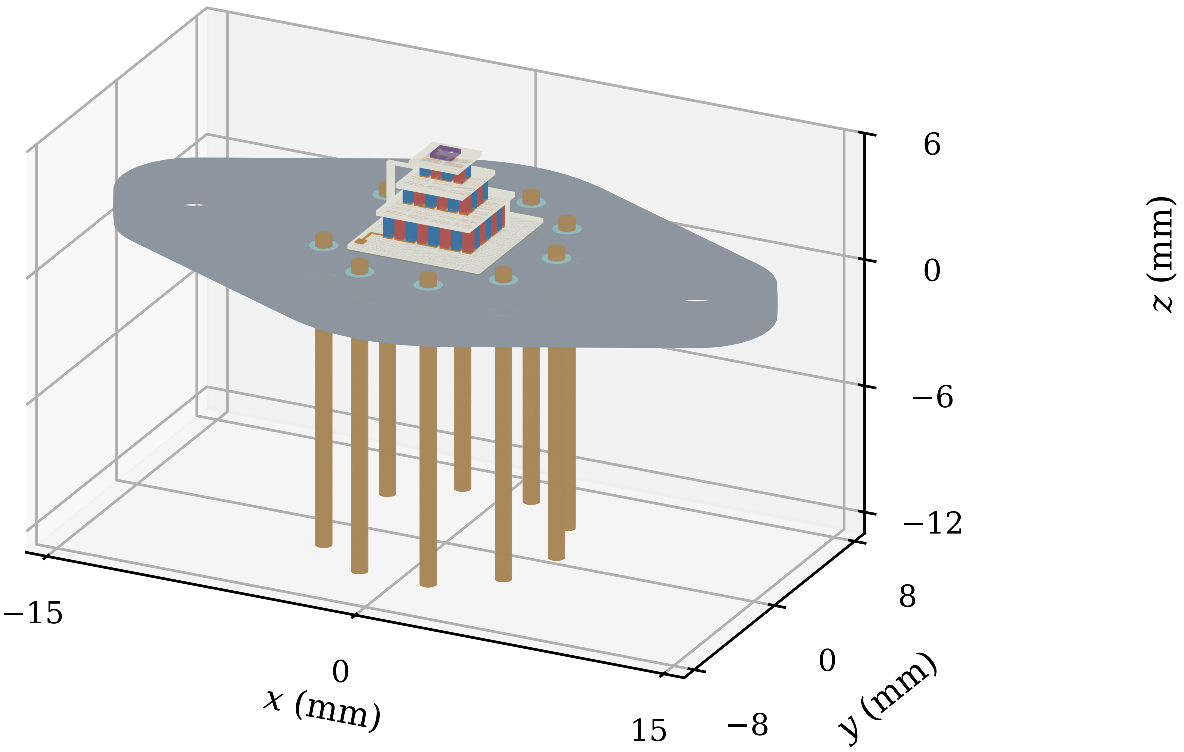

Figure 2: Actual mesh exterior: two-ear header, nine feedthrough pins and the cooler/detector stack. No cap or window is modeled. Pin geometry is retained even though the physical external electrical links are OFF.

The feedthrough seals and pin centers follow a separately audited coordinate registry. The header underside is the thermal bath; the extended pin shafts are not extra fixed-temperature boundaries. The view shows the solid domain used in the calculation, rather than a complete hermetic package.

3.1 Mesh and conductor connectivity

The solid meshes use Gmsh 4.13.1 geometry/meshing and Elmer 26.2.1 thermoelectric solution procedures. The principal field uses 159,991 nodes and 647,657 tetrahedra. Mesh checks identify material-pair contacts, exterior faces and terminal regions from actual tetrahedron adjacency, rather than inferring connectivity from a CAD rendering. No undeclared internal boundary is permitted. Every intended leg and interstage conductor belongs to the same two-terminal series component.

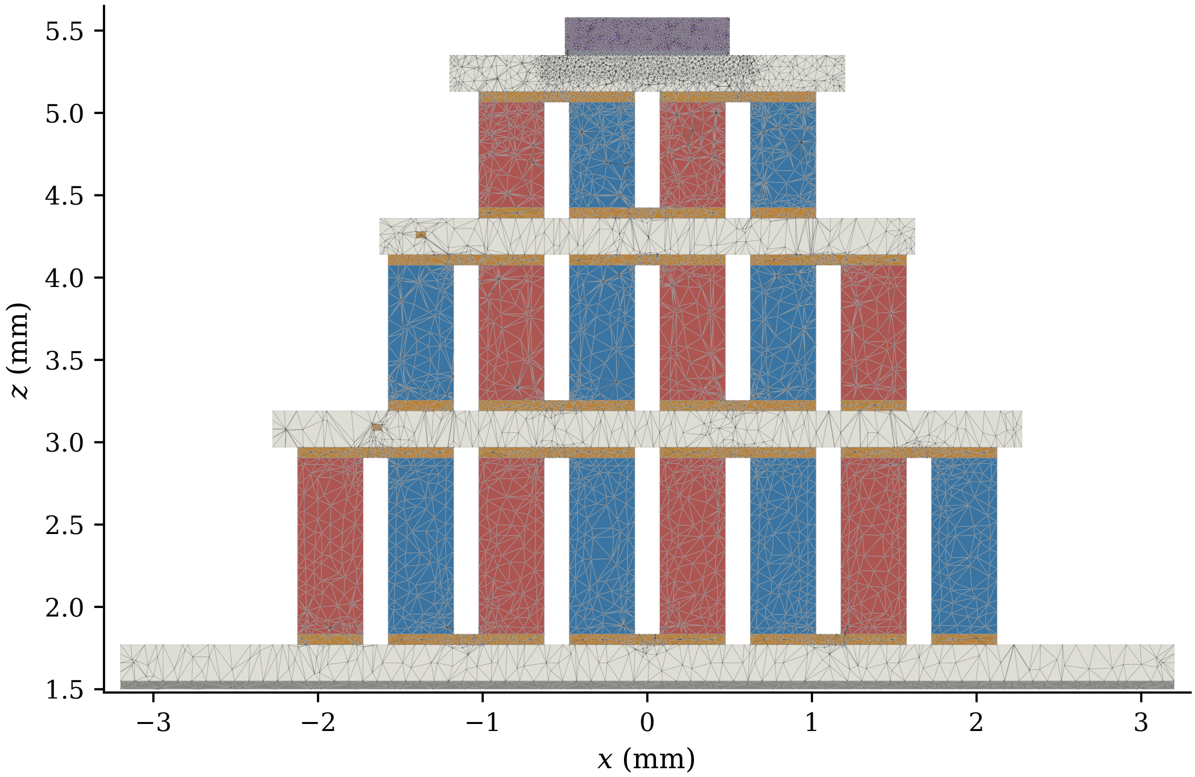

Figure 3: Exact $y = 0.15\,\text{mm}$ tetrahedron/plane intersection of the resolved stack. Void space remains white. Lines are actual cut-element edges; no scattered triangulation bridges gaps. The two hot-side module pad faces belong to the resolved copper routing. P legs are red, N legs blue, copper ochre, ceramics ivory, attach gray and die purple.

Resolved feature	Dimensions and interpretation
Ceramic plates	Widths 6.4, 4.55, 3.25 and 2.4 mm; thickness 220 μm each.
Leg sections	0.4 by 0.4 mm; stage heights 1.07, 0.82 and 0.64 mm.
Stage allocation	56, 24 and 8 legs from hot to cold; assumed internal design.
Current checks	177 interior cuts across semiconductor and copper route.
Interfaces	Shared conformal nodes; finite bulk attach and joint layers.

4 Material laws, interfaces and uncertainties

Solid	k $\mathrm{W\,m^{-1}K^{-1}}$	σ $\mathrm{S\,m^{-1}}$	S $\mathrm{\mu V\,K^{-1}}$	Qualification
P leg	1.5	140000	+160	Constant verification proxy
N leg	1.5	165500	−170	Constant verification proxy
Copper	400	5.814×10^7	0	Constant proxy
Alumina	30	—	0	Unidentified grade
Kovar/pins	17.3	—	0	Assumed alloy response
Glass	1	—	0	Seal proxy
Hot metallic joint	35	—	0	Unidentified alloy
Die attach	1	—	0	Unidentified material
HgCdTe	$k(T)$	—	0	Literature thermal model

For $\mathrm{Hg_{0.71}Cd_{0.29}Te}$, the literature model [3] is represented by piecewise linear nodal conductivity: $(T, k) = (90, 2.41)$, $(148, 1.61)$, $(298, 0.973)$ and $(400, 0.790)$ in kelvin and $\mathrm{W\,m^{-1}K^{-1}}$. The calculation stays within this range. Electrical conductivity of passive domains is a computational extension of $10^{-12}\,\mathrm{S\,m^{-1}}$, not a physical detector or pin current prediction. Density and heat capacity do not affect this steady calculation.

Interfaces have conformal shared nodes and ideal temperature/potential continuity. The explicit joint and attach have finite bulk thermal resistance; no additional calibrated contact impedance is added. Temperature-dependent, traceable P/N transport curves and measured contact resistances have *not* been qualified for this hypothetical cooler. Earlier material studies cannot identify the actual grade of an unspecified assembly.

4.1 Imposed conditions and exclusions

The electrical drive is 0.1 A with one hot-pad reference potential. Only the external Kovar underside is held at 300 K. The die receives uniform volumetric heating totaling 2 mW: 1.5 mW imposed bias dissipation plus 0.5 mW optical-like imposed heat. This does not simulate detector current, absorption, responsivity or transmission through a window.

All other exposed solid surfaces are adiabatic; no arbitrary 300 K condition is imposed on the cold assembly or pins. Copper supply and gold detector hybrid links are OFF. The source drawing's cap and window, any carrier, thermistor internals, gas, radiation and carrier-transport PDEs are excluded. These exclusions apply to this solid-only field. The separate reduced package model later includes specified wire and enclosure exchanges without creating a three-dimensional cavity field.

4.2 Temperature-dependent research analog for reduced transport

The reduced model uses source-traced P/N research-specimen curves from Zhang et al. [4]. Complete measured-table coverage is 40–300 K. The P-HE and N-HE-1 samples are distinct research specimens; they are not an identified commercial matched pair. The selected operating model restricts all leg temperatures to 180–390 K. Between 180 and 300 K, properties are interpolated on the source knots. Between 300 and 390 K, the slope of each property is *assumed* to be one quarter of its final measured-interval tangent. No measured validity above 300 K is implied.

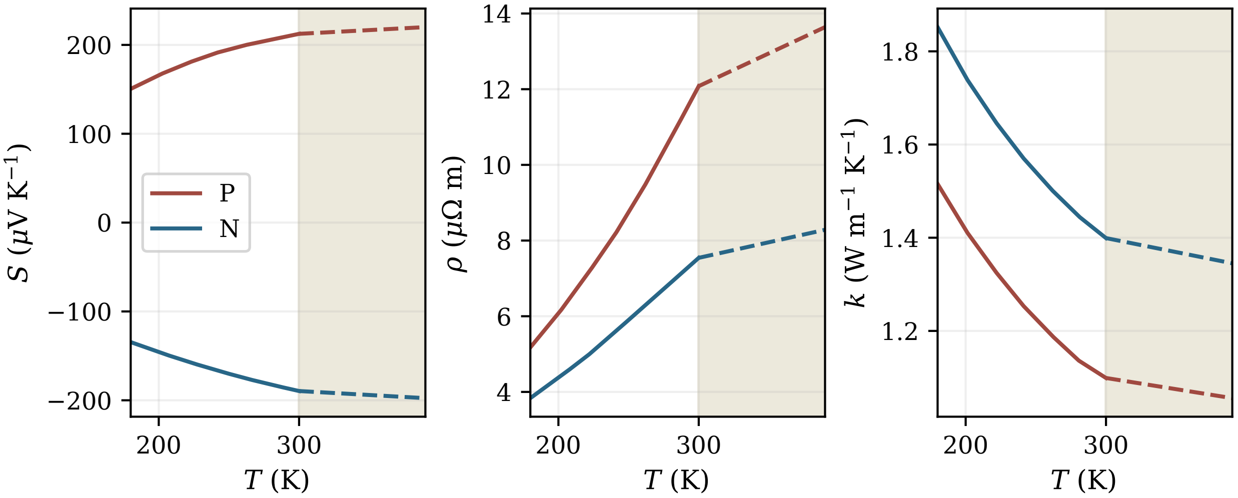

Figure 4: P/N active material laws used only by the reduced analog models. Solid lines lie within the measured-table domain; dashed lines and the shaded region denote the assumed 300–390 K continuation. The three-dimensional field shown elsewhere retains the constant verification properties.

Resistivity ρ is in $\Omega\,\mathrm{m}$, conductivity is identically $\sigma = 1/\rho$, thermopower S is in $\mathrm{V\,K^{-1}}$, and k is in $\mathrm{W\,m^{-1}K^{-1}}$. The Thomson coefficient is $\tau = T\,dS/dT$. At an interior source knot the right derivative is used; exactly 300 K uses the measured left derivative, strictly above 300 K the quarter tangent, and exactly 390 K its left limit. A two-sided derivative is not asserted at the slope discontinuity.

All evaluated states must be finite, with positive ρ and k and the correct P/N sign of S. No solution is accepted by clipping it to a temperature limit. Interior Joule heating can activate the assumed continuation even when both boundary temperatures are below 300 K. Continuation tests establish internal consistency only, not experimental validation. The research-specimen choice, digitization and assumed extension are consequential physical uncertainties.

4.3 Coherent transport hypotheses and local pair characterization

Three fixed transport triples scale the complete nominal laws, $(S, \rho, k) = (m_S S_0, m_\rho \rho_0, m_k k_0)$, for both polarities. Their source-aligned temperature dependence is retained, but the lower/higher triples are **engineering hypotheses, not measured commercial grades**. The source supplies a same-specimen triple within each polarity; it does not establish that the two specimens form a matched cooler pair [4]. No multiplier was selected by fitting manufacturer curves. Their joint covariance and manufacturability are unknown.

Hypothesis	(m_S, m_ρ, m_k)	ZT_{pair} at 200 / 300 K
Nominal	(1, 1, 1)	0.590 / 0.989
Lower	(0.9, 1.2, 1.2)	0.332 / 0.556
Higher	(1.1, 0.8, 0.8)	1.115 / 1.870

For equal-area, equal-length P/N legs, the local idealized couple indicator is

$$ZT_{pair} = \frac{(S_P - S_N)^2 T}{(\rho_P + \rho_N)(k_P + k_N)}. \tag{1}$$

Here ZT_{pair} is dimensionless and all properties are evaluated at the same T. This is neither a cascade figure of merit nor its operating coefficient of performance. Relative to nominal, the lower and higher hypotheses scale this indicator by 0.5625 and 1.890625, respectively; favorable values do not identify an available manufactured grade.

The inferred stages contain 28, 12 and 4 couples. Each leg has a 0.4 by 0.4 mm section; stage lengths are 1.07, 0.82 and 0.64 mm. It is fixed throughout the screening, as are the 32-interval nonlinear transport algorithm, source signs, reciprocal conductivity and one-sided Thomson semantics. The constant-property controls and earlier affine/nonlinear analog results are retained as separate baselines. Neither these hypotheses nor the apparent precision of their outputs resolves unknown electrical/thermal contacts, lateral ceramic spreading, current localization or buried Cu/alumina leakage.

Coverage is determined from the *entire solved leg profile*, including Joule-heated interior extrema, rather than junction temperatures alone. An accepted operating point can therefore require the assumed continuation even when its hot ceramic is nominally 300 K. The operating and manufacturer comparisons below distinguish such points from entirely measured-domain transport. Numerical acceptance of the continuation is not experimental support for its high-temperature slope.

4.4 Geometry–material scaling and structural non-identifiability

Let $a = A/A_0$, $l = L/L_0$, $g = a/l$ and let r, h, s multiply the nominal resistivity, thermal conductivity and Seebeck laws. These dimensionless factors are hypotheses; the native 88-leg dimensions remain the baseline. For one signed P/N leg current $i = \pm I$, normalized position $x = z/L$, dimensional $G = A/L$ in m and leg heat flow $F = Aq$ in W,

$$F = iST - Gk\,\frac{dT}{dx}, \tag{2}$$

$$\frac{dF}{dx} = i\left(\frac{i\rho}{G} + S\frac{dT}{dx}\right), \qquad \frac{dv}{dx} = \frac{i\rho}{G} + S\frac{dT}{dx}. \tag{3}$$

Only Gk, ρ/G and S enter. Thus the nonlinear reduced model depends on r/g, hg and s, including Joule/Thomson curvature. The transformation

$$(g, r, h, s) \longmapsto (cg, cr, h/c, s), \qquad c > 0, \tag{4}$$

leaves all steady endpoint heat/voltage and normalized leg profiles unchanged within the active domain. This is a *global non-identifiability*, independent of data precision. Common geometry preserves RK and ideal pair ZT, whereas Seebeck scaling changes ZT through $s^2/(rh)$. Near-unit rh alone cannot establish equivalence to a property triple that also changes S.

A preregistered audit attempted 269 reduced cases: 260 admitted and nine rejected; 71 admitted cases stay inside measured coverage and 189 use continuation. All 35 admitted equivalence checks pass, with maximum temperature difference below 10^{-9} K; one pair cannot be compared because its solves reject. No fitting occurs. Separate A and L also remain ambiguous in this isolated steady model: scaling both together changes spatial flux/current densities while preserving total powers and normalized profiles. Real spreading/contact geometry is absent.

Log step	σ_1	σ_2	σ_3	$10^5\sigma_4$	Raw rank
0.005	4.733125	1.943249	0.594098	5.18915	4
0.0025	4.733123	1.943067	0.594094	1.05019	3

Here rows are lift/100 K and voltage/1 V at six registered 300.15 K conditions; columns are log g, r, h, s. The rank threshold is $10^{-5}\sigma_1$. Step halving changes the matrix by $5.23\,10^{-5}$ in relative Frobenius norm, but the coarse fourth singular value barely clears the threshold. Raw rank 4 is therefore not evidence against the exact null direction $(1, 1, -1, 0)$. Its fine-step cosine exceeds 0.99999999997 with that direction; retained conditioning is 7.967. Measured-domain 295 K controls also give rank 3. These local ranks do not prove uniqueness of the remaining effective laws or provide a likelihood/confidence interval. Independently known dimensions or transport/contact evidence are needed to distinguish the aliased factors.

5 Coupled field equations and solution verification

With electric potential ϕ, $\mathbf{E} = -\nabla\phi$, conductivity σ, thermal conductivity k and Seebeck coefficient S, the steady constitutive equations are

$$\mathbf{J} = \sigma(-\nabla\phi - S\nabla T), \qquad \nabla\cdot\mathbf{J} = 0, \tag{5}$$

$$\mathbf{q} = -k\nabla T + ST\mathbf{J}, \qquad \nabla\cdot\mathbf{q} = \mathbf{J}\cdot\mathbf{E} + q_v. \tag{6}$$

The coupled upstream Elmer thermoelectric solver includes Seebeck, Peltier, Joule and Fourier terms. It uses conformal linear tetrahedra and a nonlinear stopping threshold below 10^{-9}. The reduced models introduced later are separate calculations; they do not replace or relabel this three-dimensional field.

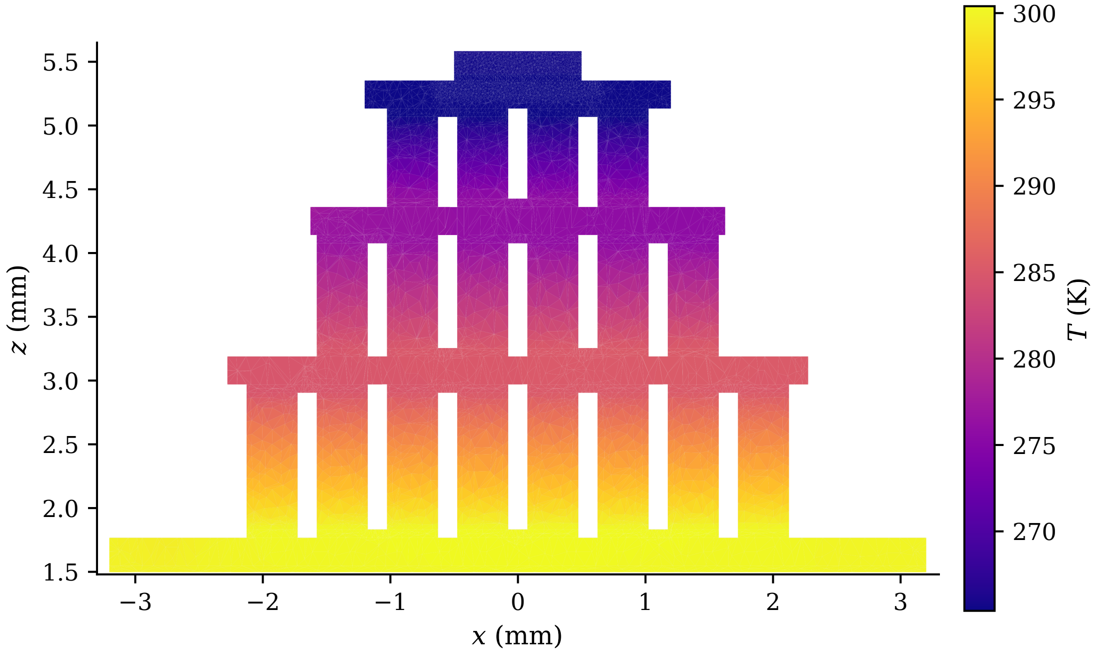

Figure 5: Actual solved temperature on the exact stack section. Colors represent source-cell means of the P1 field. The colder upper stage and warmer hot-side solids arise from the coupled solution. The field is conditional on the assumed materials and failed spatial qualification.

Principal archived calculation	Numerical value
Nodes / tetrahedra	159,991 / 647,657
Die volume mean	265.616093 K
Cold ceramic volume mean	265.424850 K
Terminal voltage	0.551926633 V
Known input, $I\Delta\phi + Q_{die}$	57.192663 mW
Bath weak reaction / raw gradient	57.192663 / 56.336914 mW

Extra digits identify reproducible numerical outputs; they do not express physical accuracy or manufacturing tolerance.

6 Electrical and heat-flow fields

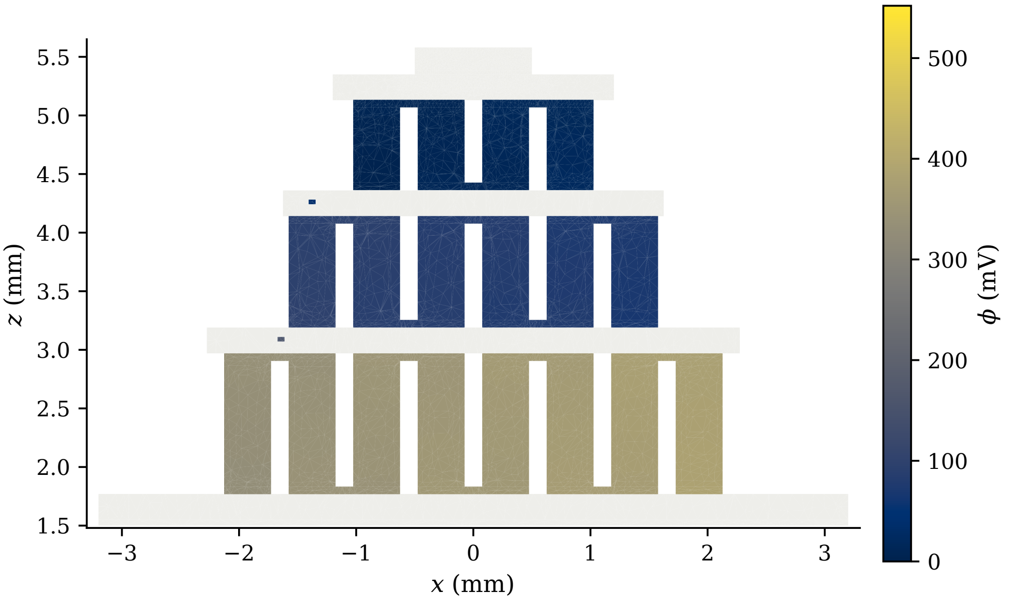

Figure 6: Actual electrical potential on the conductor section. Only P, N and copper are colored. The module terminals are on the hot side; external pin links are OFF.

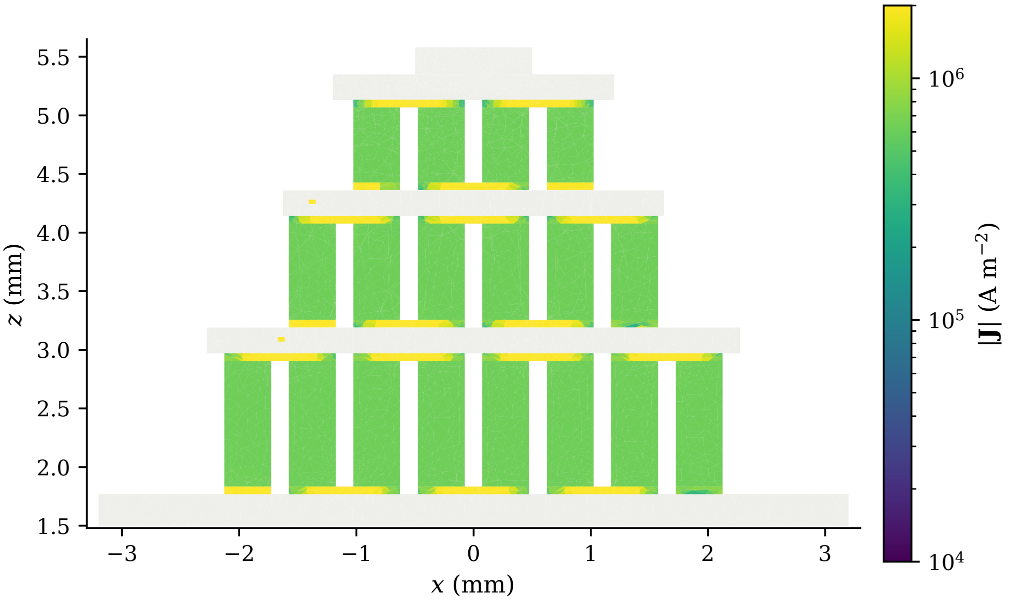

Figure 7: Magnitude of the actual element-mean current density on the same section. Continuity is tested separately at every one of 177 interior cuts; plotting a smooth field alone is not a continuity test.

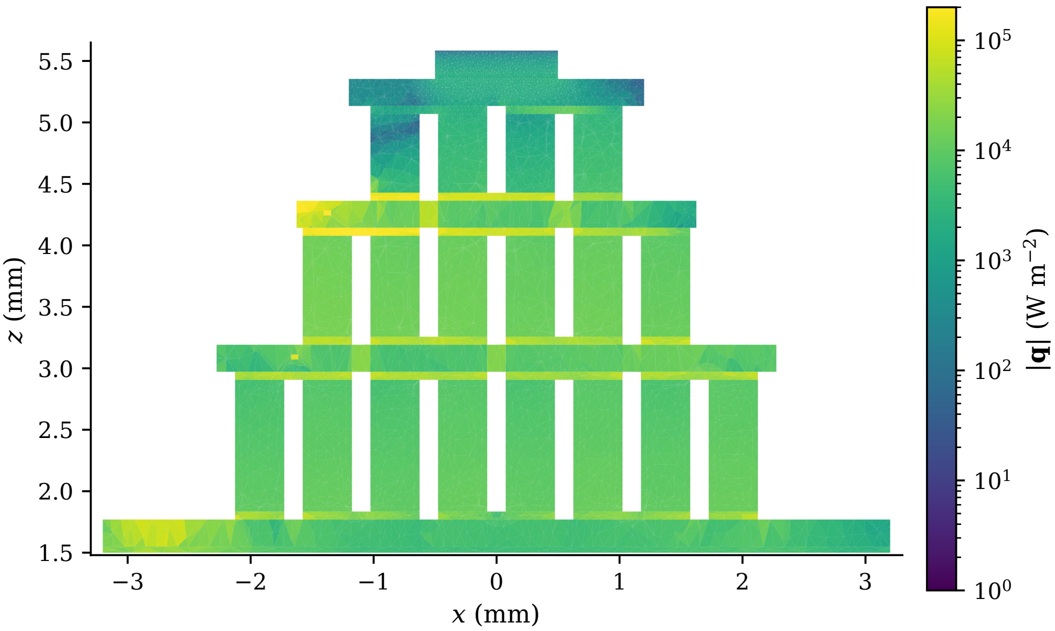

Figure 8: Actual element-mean total solid heat flux, including Fourier and Peltier terms. The logarithmic scale reveals both the stack and passive conductive paths; it is not a recovered conservative boundary flux.

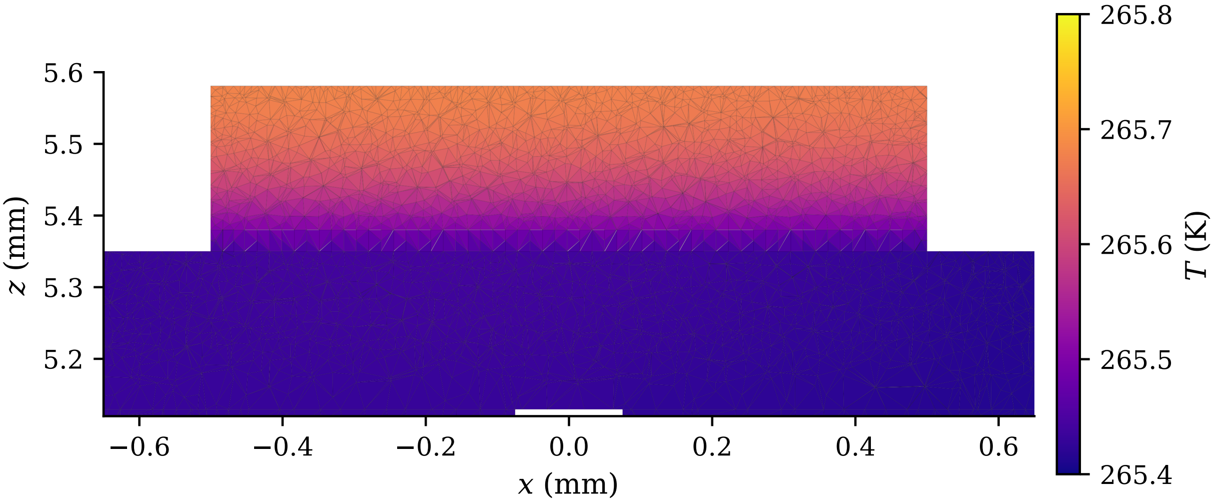

Figure 9: Actual resolved cold ceramic, 30 μm attach and 200 μm die section. The small die-temperature mesh sensitivity does not certify the normal heat-flux gradients. The maximum plotted range is a visualization choice, not an uncertainty band.

7 Independent raw-gradient diagnostics

The heat-flux audit fits each tetrahedron's actual affine temperature gradient, uses its owner material conductivity and the oriented area of each physical triangle, and sums powers with area weights. At an internal interface, each solid supplies an independent outward trace. Define

$$e_\Gamma = \frac{Q_{out}^- + Q_{out}^+}{(|Q_{out}^-| + |Q_{out}^+|)/2}, \qquad e_b = \frac{Q_{b,raw} - (I\Delta\phi + Q_{die})}{I\Delta\phi + Q_{die}}. \tag{7}$$

The unchanged limits are $|e_\Gamma| < 5\%$ and $|e_b| < 1\%$. Reactions and recovered fluxes are useful checks but cannot replace these raw criteria.

8 Cold-interface and bath evidence

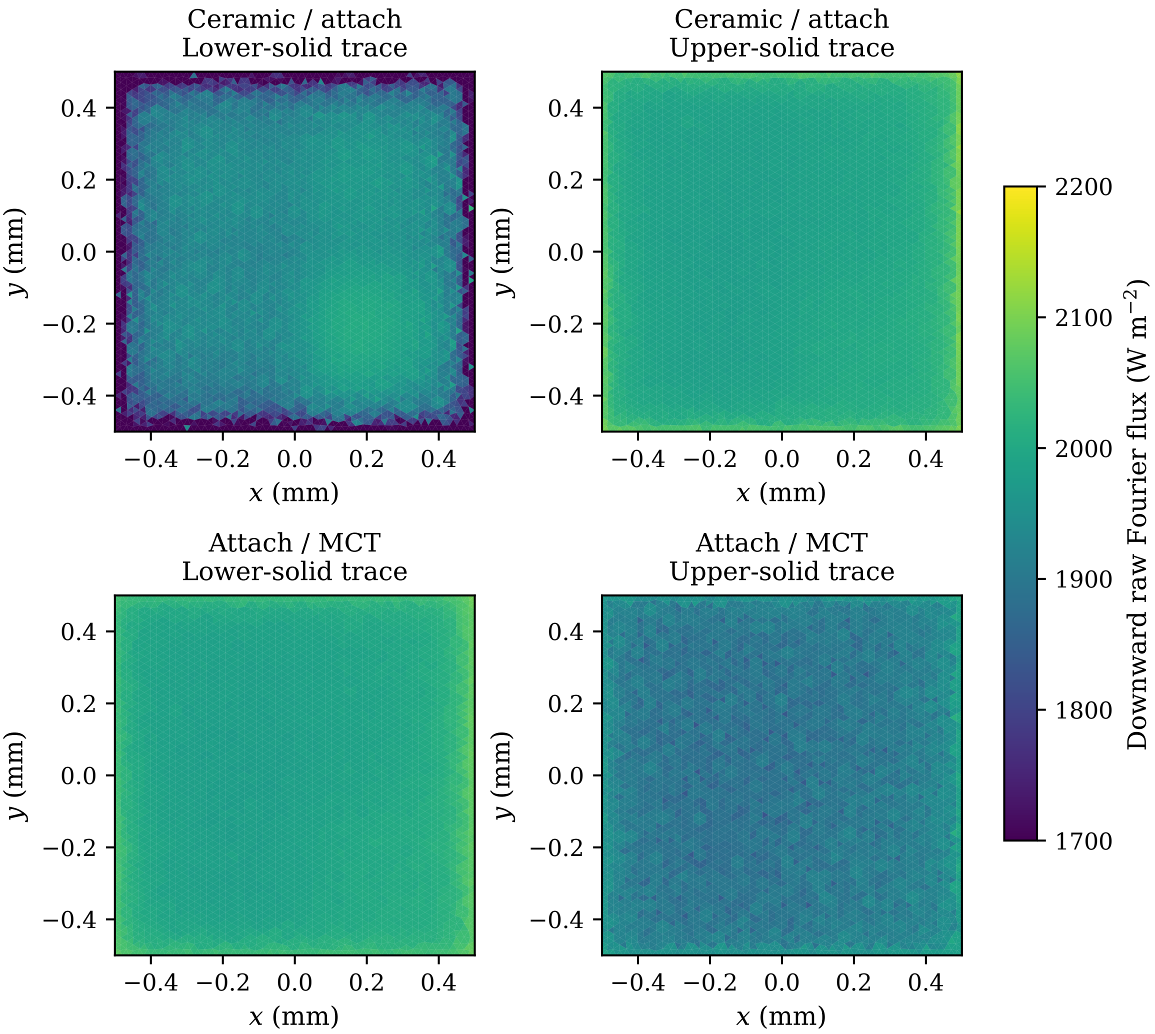

Figure 10: Actual independent raw Fourier flux on both sides of each cold contact, positive downward. The ceramic/attach mismatch is +6.4765% (FAIL); attach/MCT is −4.6085% (PASS). Different traces are numerical evidence, not an imposed contact jump.

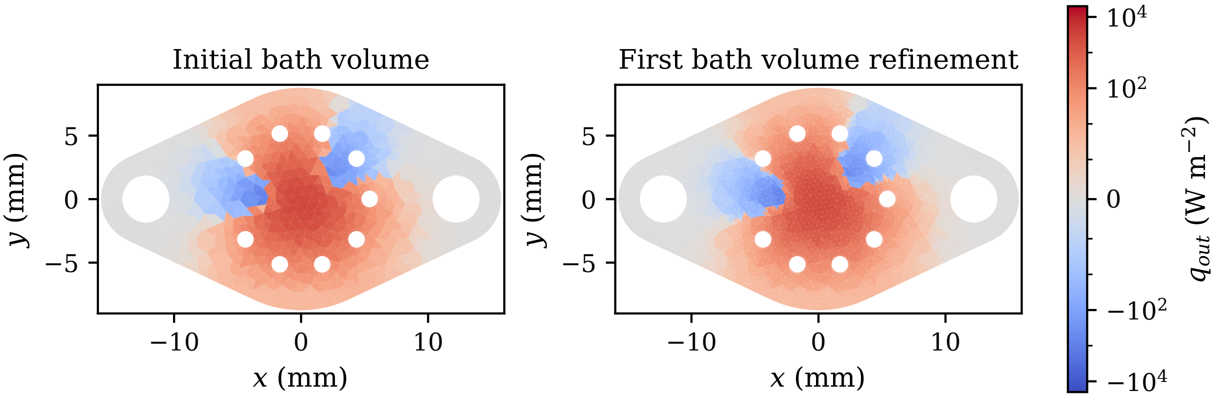

Figure 11: Actual raw outward flux on the Kovar underside before and after a separate bath-volume refinement. The cold mesh is exactly fixed. Absolute integral errors improve from 1.4963% to 1.2914%, both FAIL. The reference is known external input, not the bath reaction.

9 Mesh sensitivity and independent analytical benchmark

Mesh	Tetrahedra	Die mean (K)	Voltage (V)	$e_{cer/att}$ (%)	$e_{att/die}$ (%)
A	366,156	265.592524	0.551939871	+41.348	-47.828
B	458,061	265.613622	0.551945058	+8.695	-6.976
C	647,657	265.616093	0.551926633	+6.477	-4.609

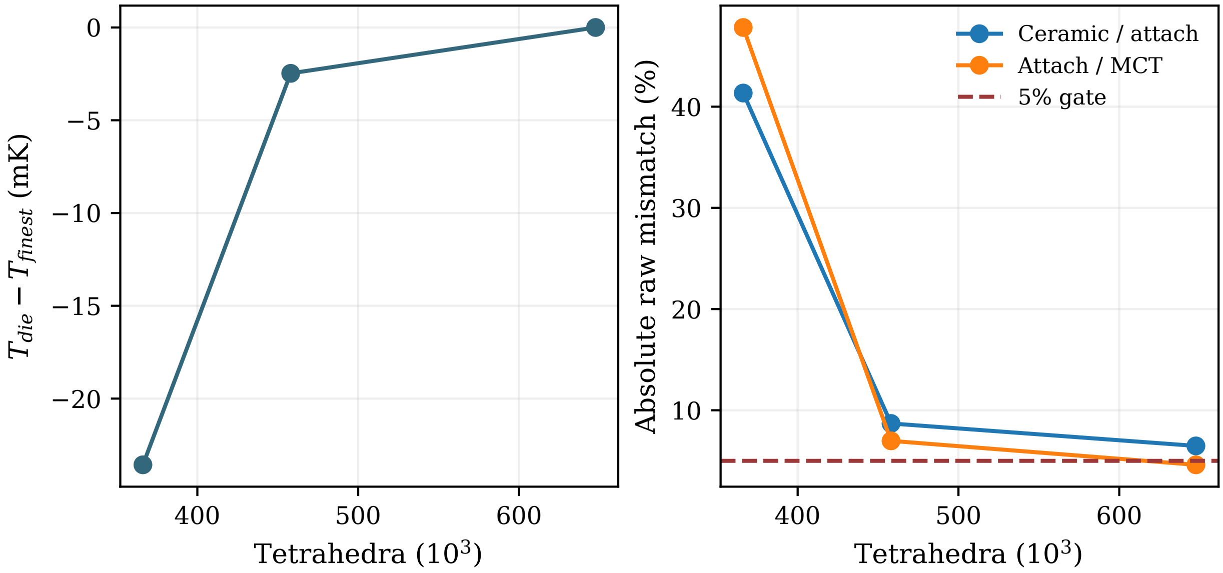

Figure 12: Three actually completed coupled meshes. Successive die-temperature changes are 21.10 and 2.47 mK; voltage is nonmonotone. Raw ceramic/attach error remains above 5%. These are sensitivity data, not a rigorous whole-device grid-convergence index.

The cold-region effective mesh sizes are 76.724, 25.484 and 17.608 μm. Refinement is local and anisotropic; the bath and most TEC regions stay nearly fixed. A conditional temperature-only fit yields order 1.324 and a 3.91 mK fine-to-limit difference, but this does *not* qualify a device GCI. Nonmonotone voltage admits no such extrapolation. The separate 675,496-tetrahedron bath case shifts the die mean by +0.421 mK and voltage by +2.946 μV.

9.1 Independent analytical reference

The benchmark uses 220/30/200 μm layers and conductivities 30/1/1.1 W m^{-1}K^{-1}. Constant-flux and distributed-source cases separate normal-gradient truncation from integration errors. Coarser benchmark failures remain failures. Near-exact nodal values with finite mean and face-flux errors illustrate why an algebraically converged solution may still fail raw-gradient gates.

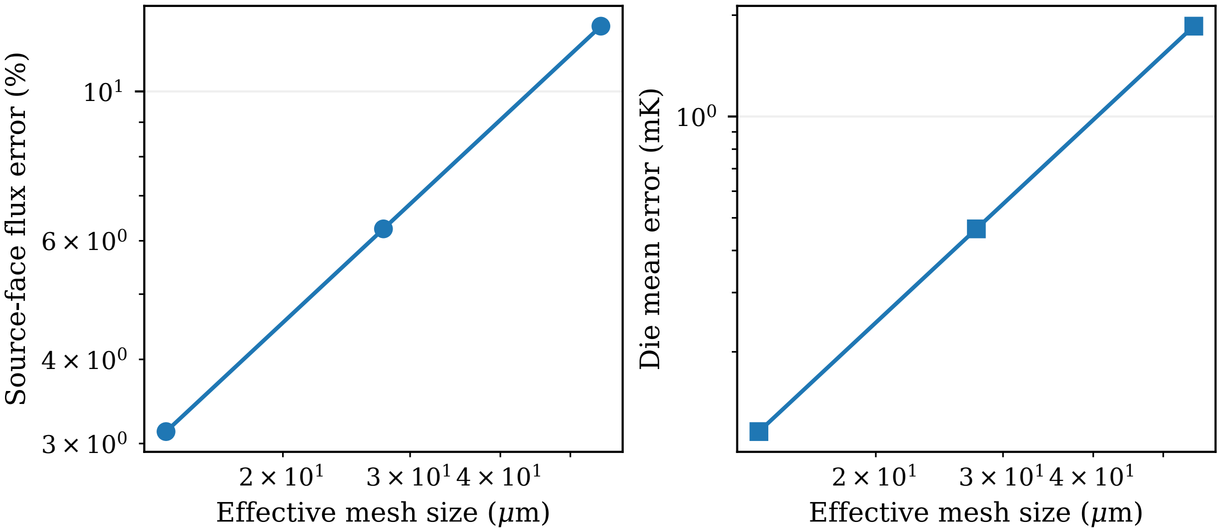

Figure 13: Independent three-dimensional, three-layer HeatSolver benchmark with an analytical temperature field, harmonic transverse variation and a distributed 2 mW source. Source-face errors are 12.5, 6.25 and 3.125%; mean-temperature errors are 1.855, 0.464 and 0.116 mK. The finest benchmark passes its benchmark-specific preregistered checks; these do not replace the production bath criterion or qualify the coupled device.

For a uniformly heated slab with constant conductivity, the axial solution includes a quadratic term, $T(z) = a + bz - q_v z^2/(2k)$. Its exact gradient varies linearly. A linear tetrahedron represents temperature by an affine field and therefore has a constant gradient. Even when nodal temperatures nearly match the analytical solution, the boundary trace samples a finite owner volume rather than the exact boundary derivative. Normal refinement addresses that truncation directly.

The analytical mean temperature is 300.1958788 K. The finest distributed-source calculation gives 300.1957629 K. Source-face flux error halves at each level, while mean-temperature error decreases by approximately four, consistent with first-order boundary gradients and second-order temperature integration in this controlled benchmark. The actual coupled device has material jumps, geometric edges and local refinement, so these rates are evidence for the mechanism, not transferable error estimates.

10 Verification matrix and numerical release gates

This matrix concerns the illustrated three-dimensional solid field. Independent refinement sequences are evaluated separately below; a reduced-model balance cannot certify a spatial flux gate.

Test	Hard limit	Baseline evidence	Status
Source geometry/topology	216 domains; 88 legs; two terminals	Corrected actual mesh audit	PASS
All 177 current cuts	Each $< 0.5\%$	Worst 0.406120%	PASS
Global heat residual	$< 1\,\mu$W	$0.000144\,\mu$W	PASS
Free thermal residual	$< 1\,\mu$W	$0.001838\,\mu$W	PASS
Free current residual	$< 1\,\mu$A	$0.042206\,\mu$A	PASS
Nonlinear change	$< 10^{-9}$	$4.34\,10^{-10}$	PASS
Homogeneous solid test	Passive 300 K	Error below 4 nK	PASS
Same-mesh replay	Agreement with native LU	$0.225\,\mu$K; 3.70 nV	PASS
Ceramic/attach raw flux	$< 5\%$ absolute	6.48%	**FAIL**
Attach/MCT raw flux	$< 5\%$ absolute	4.61%	PASS
Independent raw bath	$< 1\%$ absolute	Best normal case 1.069%	**FAIL**
Same production mesh and refined comparator	Both raw gates, all conservation checks	Not demonstrated	OPEN
Qualified whole-device GCI	Systematic asymptotic evidence	Local conditional fit only	OPEN
Empirical device accuracy	Independent measurements	No measured reference article	OPEN

Numerical release requires both original raw limits on one production mesh and at least one systematically refined comparator, together with all 177 current cuts, global/stage heat checks, free thermal/electrical residuals, homogeneous baseline and reproducibility checks. Temperature and voltage sensitivity must be reported. Preferred 4.5% cold and 0.9% bath margins are targets, not relaxed or replacement gates. Passing two different meshes separately is insufficient.

11 Independent normal and transverse refinement evidence

The cold and bath refinement tracks retain the same 0.1 A, 2 mW, 300 K underside boundary and disabled external links. Their equations and raw acceptance criteria are unchanged. Free owner-volume nodes are refined with region locks; changes to a temperature output do not substitute for flux checks.

Study	Tetrahedra	Die mean (K)	Voltage (V)	Cold raw (%)	Bath raw (%)
Cold normal	821,078	265.616118	0.551926634	+4.795	-1.496
Bath normal 1	744,713	265.616580	0.551930113	+6.477	-1.069
Bath normal 2	821,458	265.616593	0.551930210	+6.477	-1.093

The cold-only track brings the ceramic/attach raw mismatch to 4.795%, passing the original 5% gate on that mesh; its bath still fails. The contact above it improves to 2.306%. Die mean changes only 24.94 μK while a boundary gradient changes materially. The bath-only normal-height sequence is approximately 296, 148 and 74 μm; its 1.069% then 1.093% errors are nonmonotone, both FAIL. Separate tracks do not establish a same-mesh release.

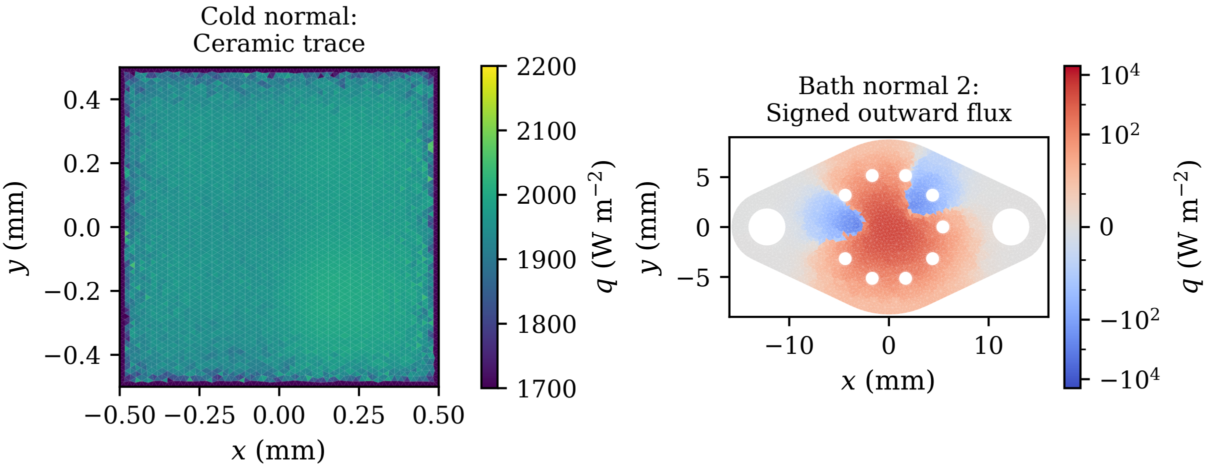

Figure 14: Actual cold-normal ceramic trace and second bath-normal signed outward trace, from separate genuine solved fields. The bath map belongs to the 821,458-tetrahedron normal case. It is not the later transverse field and does not depict a combined passing mesh.

11.1 Stage conservation and later bath-volume behavior

The second bath-normal field provides the following independent weak control-volume decomposition. The coldest band includes the 2 mW die source, so its electrical-work difference alone is not the whole band balance.

Stage band	Hot out (mW)	Cold out (mW)	Electrical work (mW)	Residual (μW)
1	57.193021	18.014934	39.178087	0.000008
2	18.014934	5.754100	12.260833	-0.000018
3	5.754100	0	3.754100	-0.000006

All three normal cases pass all 177 interior current cuts, global/stage heat and free-residual checks. The worst current-cut error is 0.406120%. Maximum global heat error is 0.000093 μW; maximum free thermal and current residuals are 0.002472 μW and 0.042254 μA. Those numerical checks remain separate from raw-gradient qualification.

Two additional transverse/volume bath refinements also solve the genuine coupled solid equations. Their cold/electrical regions remain fixed. Each passes the unchanged nonraw checks, but neither passes the original bath criterion:

Study	Tetrahedra	Die mean (K)	Voltage (V)	Bath raw (%)
Transverse volume 1	784,696	265.616563	0.551929953	-1.2885
Transverse volume 2	1,099,293	265.616574	0.551930034	-1.2907

The die-temperature difference between these two fields is approximately 11 μK while bath errors plateau near 1.29%. Signed local flux regions of opposite sign and internal gradient jumps matter; global cancellation can coexist with unresolved local traces. No recovered or reaction-based flux replaces the raw bath criterion. A combined cold-plus-bath passing production mesh and a systematically refined passing comparator are absent.

11.2 Computational method and practical cost

The fixed fine solid problem was solved with a native direct factorization and checked with a same-mesh numeric-refinement replay. The principal native calculation takes approximately 1,292 s and 5.7 GiB peak resident memory; a warm replay completes in approximately 558 s and 5.8 GiB. This is a warm-start comparison, not a general cold-start speed guarantee. The later coupled transverse fields complete in approximately 911 and 999 s with three nonlinear iterations. These numerical costs motivate bounded transport/exchange reductions for engineering scenarios, without implying that the reduced model reproduces the full spatial boundary-value problem.

12 Conservative nonlinear transport and five-temperature package

The reduced package has unknown temperatures T_h, T_1, T_2, T_c, T_d: hot ceramic, two intermediate ceramics, cold ceramic and die. Each stage has a representative P leg and N leg, scaled by its assumed number of identical couples. No current-spreading field or per-leg manufacturing variability is solved.

Let z run from hot to cold, L be leg length and A its area. Current densities are $j_N = I/A$ and $j_P = -I/A$. The steady one-dimensional laws are

$$q = STj - k\,dT/dz, \qquad dq/dz = j(\rho j + S\,dT/dz), \tag{8}$$

$$v_s = j\rho(T_m)\Delta z + S(T_m)\Delta T, \qquad w_s = jv_s, \tag{9}$$

$$r_i = q_{i+1/2} - q_{i-1/2} - \tfrac{1}{2}(w_{i+1/2} + w_{i-1/2}) = 0. \tag{10}$$

Here q is $\mathrm{W\,m^{-2}}$, v_s is V, w_s and r_i are $\mathrm{W\,m^{-2}}$, T_m is segment-midpoint temperature and $\Delta z = L/n$. Endpoint heat fluxes include the half-segment work. Both endpoint heats are independently computed:

$$\begin{aligned} Q_c &= -NA(q_{c,P} + q_{c,N}) - K_b(T_h - T_c), \\ Q_h &= -NA(q_{h,P} + q_{h,N}) - K_b(T_h - T_c), \end{aligned} \qquad V = N(v_N - v_P). \tag{11}$$

K_b is an explicit $\mathrm{W\,K^{-1}}$ bypass hypothesis, zero in these scenarios. The identity $Q_h - Q_c = IV$ is audited rather than imposed by overwriting a flux. Variable thermopower already contributes through flux and electrical work; an extra Thomson heat source would double-count it.

An analytic tridiagonal Jacobian and banded Newton solve use a residual-decreasing bounded line search. Nominal legs have 32 intervals. Material domains are enforced at all evaluations. Previously evaluated stage problems are reused only for identical dimensions, boundary temperatures, current and material multipliers; inputs are not rounded. A constant-property package solution supplies an initial iterate only. Failed nonlinear/residual/domain checks remain rejected states with recorded candidate diagnostics.

12.1 Package, wire and load balances

At each intermediate node, preceding-stage Q_c balances the next Q_h plus net parasitic heat. The cold ceramic balances stage cooling against die-attach and local exchange; the die balances imposed heating, bond conduction and enclosure exchange against attach heat:

$$Q_{c,0} = Q_{h,1} + E_1, \qquad Q_{c,1} = Q_{h,2} + E_2, \tag{12}$$

$$Q_{c,2} = E_c + G_a(T_d - T_c), \qquad G_a(T_d - T_c) = E_d + Q_{bias} + Q_{abs}. \tag{13}$$

E is heat into the corresponding node in W; $G_a = k_a A_a / t_a$, nominally $1/30\,\mathrm{W\,K^{-1}}$. The effective hot-side condition is

$$T_h - T_{amb} = R_{ha}\{Q_{h,0} + Q_{Cu,cond} + \tfrac{1}{2}I^2 R_{Cu} + E_h\}. \tag{14}$$

$R_{ha} = 2\,\mathrm{K\,W^{-1}}$ nominally represents the full hot-side path, not a resolved joint/header/spreader resistance. Corrected Cu routes join pins 1/9 to *hot pads*; Au routes join pins 7/8 to the detector. Conductances are kA/L and electrical resistances $\rho L/A$. Cu Joule heat is split between its warm pin and hot pad; terminal voltage includes that series loss. Bias and absorbed heat are prescribed, not inferred from an optical or carrier calculation.

12.2 Independent transport accuracy and refinement

Eight representative boundary-value cases keep all reference temperatures strictly inside the measured 40–300 K interval. Two additional cases test the assumed high-temperature law across implementations only. The independent bounded nonlinear reference uses the same material tables and conservative midpoint equations but a different bounded least-squares algorithm. Same-discretization agreement checks implementation; 16/32/64 interval comparisons probe discretization, not physical material accuracy.

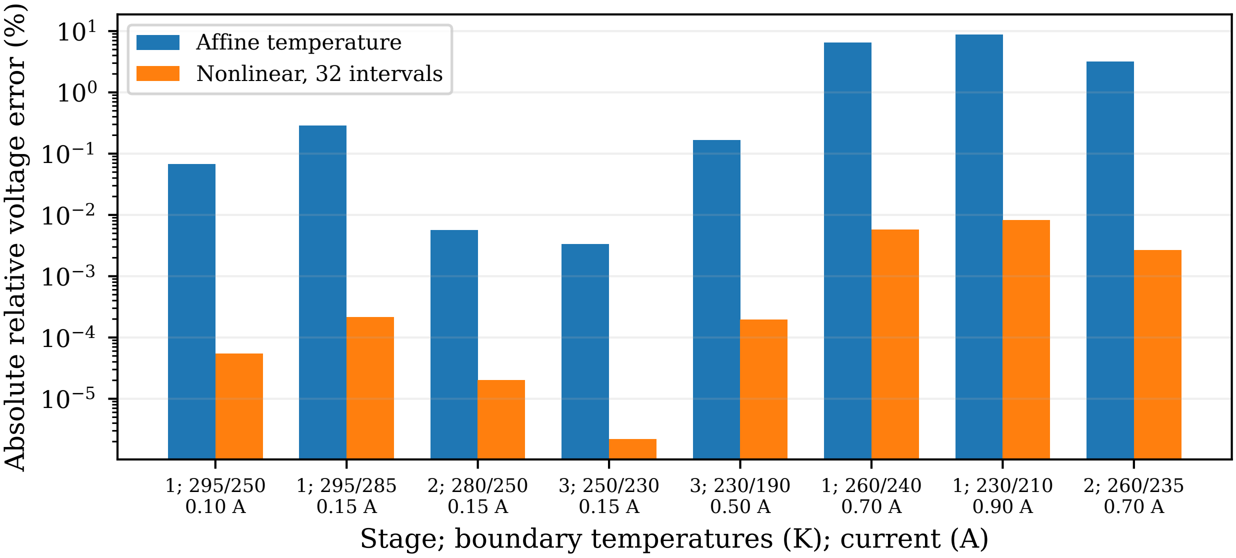

Figure 15: Absolute relative stage-voltage error against the independent 64-interval nonlinear reference. Stage numbering is hot to cold. The affine-temperature assumption omits Joule/Thomson-driven profile curvature; the direct nonlinear calculation substantially reduces that approximation error. These are numerical comparisons using research-specimen laws, not measurements of a device.

At $T_h/T_c = 260/240$ K and 0.7 A, nonlinear32 differs from the refined reference by 6.762 μW in Q_c (0.000823%) and 0.1294 mV (0.00570%). The earlier affine voltage difference was about 147 mV (6.48%) against the 32-interval nonlinear reference. At 230/210 K and 0.9 A, nonlinear32 voltage error is 0.00823%. The profiles stay bounded; interior temperature curvature, not a tuned coefficient, produces the improvement.

For the nominal package at 0.9 A with the older effective-reservoir closure, nonlinear32-to64 changes die temperature by 0.000350 K and terminal voltage by 0.1643 mV. Knot crossings can cause small nonmonotone changes, so no whole-device asymptotic order or grid-convergence index is inferred. Median inner transport compute time over accepted scenario points is about 0.5 s in the bounded four-CPU environment, excluding some initialization and outer-iteration overhead. Reusing identical stage evaluations reduces cost without introducing a surrogate interpolation error.

13 Candidate enclosure and preliminary exchange geometry

The candidate cap/roof/window are native solids using the documented cap outside diameter and explicit engineering hypotheses for wall, height and window dimensions. Cap height above the header is 8.60 mm, wall 0.50 mm, roof/window thickness 0.45 mm and window diameter 5.90 mm. The inner roof is at $z = 9.65$ mm, approximately 4.07 mm above the die top. Steel-like cap and ZnSe-like window descriptions are candidate material concepts, not identified manufactured grades; no resolved window thermal/optical law is used.

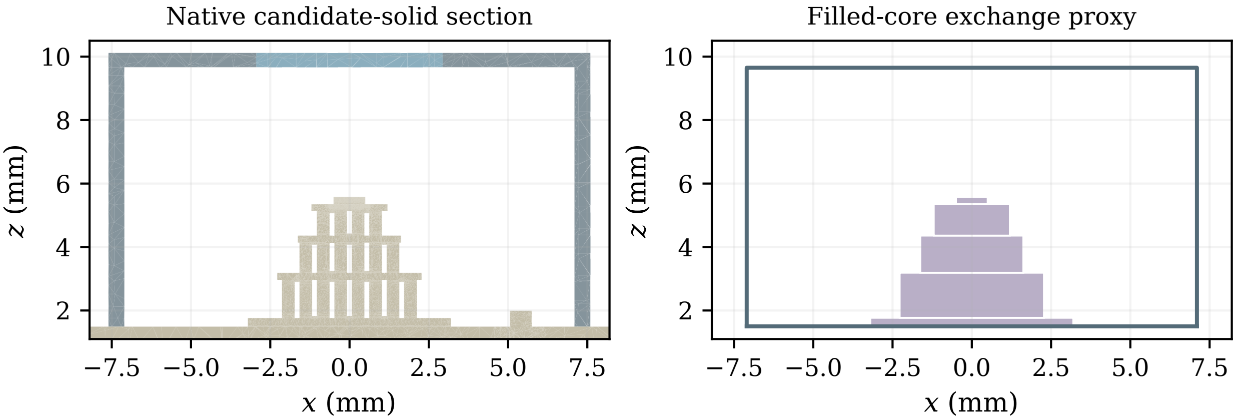

Figure 16: Left: central $y = 0.15$ mm sections of native candidate cap/window tetrahedra and the corrected solid stack, with geometric voids retained; the flange extends outside this view. The native enclosure is a separately meshed fixture, not a conformal addition to the coupled solid field. Right: the explicitly filled stepped-core geometry used only by the preliminary exchange operator. The native surfaces and proxy surfaces are different models. Neither panel is a newly solved enclosure temperature field.

The native fixture has 8,143 tetrahedra and distinct shell, roof and window volumes. Positive tetrahedron volumes, unique exterior ownership, all nine physical face groups and shell/roof and roof/window interface incidence are audited. The package exterior is independently checked against the corrected solid mesh. Native exposed ceramic-top areas are 29.964, 16.1295, 8.9375 and 4.760 mm^2; the die exposes approximately 0.955 mm^2. These are not silently substituted for the different proxy areas.

The closed exchange geometry is an inner cylinder of radius 7.1 mm with header floor and a filled stepped core based on the corrected widths and heights. Fourteen surface groups map to five unknown temperatures plus prescribed header, cap and window reservoirs. Vertical core groups use means of adjacent node temperatures. Actual interleg voids, exposed metallization, pin and wire occlusion are omitted. Thus the operator supplies a bounded, auditable preliminary load budget; full native-surface view factors remain unimplemented.

13.1 Gray radiation and pressure-dependent gas network

For gray opaque surface i, area A_i, emissivity ϵ_i, radiosity J_i and view factor F_{ij},

$$A_iF_{ij} = A_jF_{ji}, \qquad \sum_j F_{ij} = 1, \tag{15}$$

$$J_i = \epsilon_i\sigma_{SB}T_i^4 + (1-\epsilon_i)\sum_j F_{ij}J_j, \qquad Q_{rad,i} = A_i\left(\sum_j F_{ij}J_j - J_i\right). \tag{16}$$

$\sigma_{SB} = 5.670374419\,10^{-8}\,\mathrm{W\,m^{-2}K^{-4}}$. Heat is positive into a surface. Deterministic scrambled low-discrepancy area/cosine-direction rays terminate on the first opaque proxy surface. Parallel upward faces, coplanar roof/window faces and convex self-views remain blocked. The concave wall can view itself. Raw exchange areas are symmetrized and symmetrically balanced for reciprocity and closure, retaining pairs unseen in both directions as zero. Both raw and corrected matrices are retained; conservation does not conceal the correction.

The gas path is a passive *chord network*, not a solved diffusion or rarefied-gas field:

$$G_{ij} = \frac{k_g(T_{ij})A_iF_{ij}}{\ell_{ij} + 2\delta_{ij}}, \qquad Q_{g,i} = \sum_j G_{ij}(T_j - T_i), \tag{17}$$

$$\delta_{ij} = \frac{2-a}{a}\frac{2\gamma}{\gamma+1}\frac{\lambda}{Pr}, \qquad \lambda = \frac{\mu}{p}\sqrt{\frac{\pi R_g T_{ij}}{2}}, \tag{18}$$

$$\mu = \frac{k_g}{c_p + 1.25R_g}, \qquad Pr = \frac{\mu c_p}{k_g}. \tag{19}$$

$T_{ij} = (T_i + T_j)/2$ is the pair mean, $\gamma = c_p/c_v$ is the heat-capacity ratio, and Pr is the dimensionless Prandtl number. ℓ_{ij} is a symmetric area-weighted ray chord in m, G is $\mathrm{W\,K^{-1}}$, a is accommodation (assumed 0.8), p is Pa, R_g and c_p are $\mathrm{J\,kg^{-1}K^{-1}}$, and μ is Pa s. The viscosity relation is an Eucken engineering closure, not independent measured viscosity. At $p = 0$, gas exchange is exactly zero; at small pressure it scales approximately with p. At large pressure it tends to the stated Fourier *network* limit. Neither limiting check validates the network's physical geometry approximation.

The NIST gas conductivity tabulation [5] supplies N_2/Ar/Kr values at 200, 300 and 400 K. At 300 K, k_g is 0.0259, 0.0177 and 0.0095 $\mathrm{W\,m^{-1}K^{-1}}$ respectively. Interpolation is bounded; 180–200 K uses an explicitly flagged $T^{0.75}$ engineering continuation. No accurate accommodation, mixture/Xe transport, natural convection or production backfill optimum is established. The window is internally opaque gray; imposed absorbed heat is not a prediction of transmission.

13.2 Ray refinement and conservative coupling

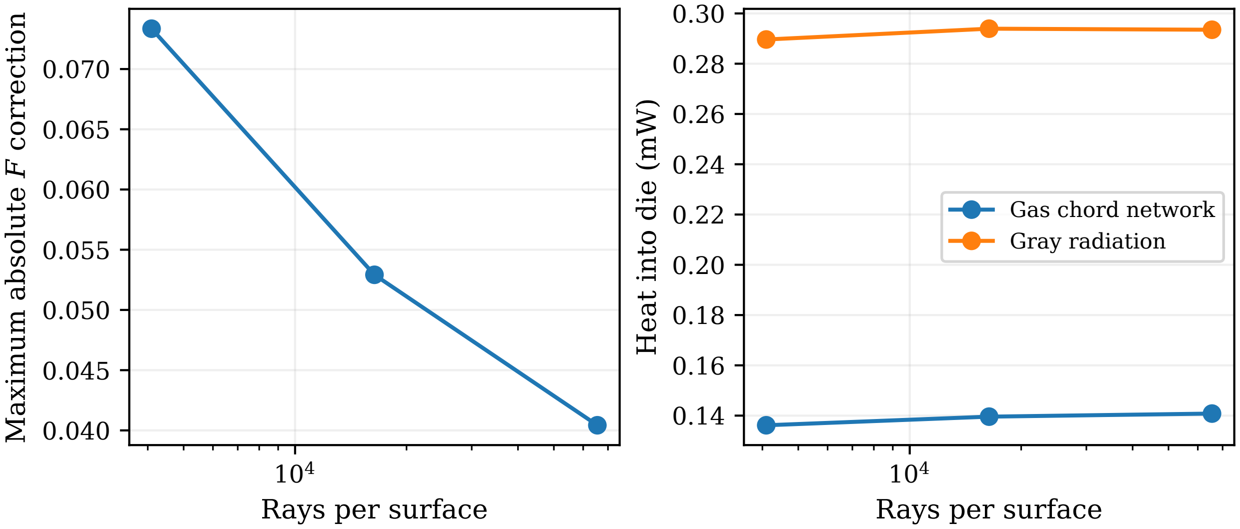

Figure 17: Bounded proxy-ray refinement at fixed package temperatures 300/270/240/220/221 K and 300 K reservoirs. Left: maximum absolute correction applied to a view factor during reciprocal closure. Right: resulting die gas and radiation heat input. These are deterministic refinement diagnostics, not statistical uncertainty bands or native-cavity error estimates.

From 4,096 to 16,384 to 65,536 rays per surface, the largest absolute view-factor corrections are 0.07335, 0.05292 and 0.04043. The last value is still a 4.04-percentage-point correction. Die gas inputs are 0.136161, 0.139590 and 0.140812 mW, and radiation inputs 0.289628, 0.293933 and 0.293511 mW. Last-level changes are approximately 0.88% gas and 0.14% radiation. This does not qualify spatial view-factor accuracy; the filled-core model-form error is not estimated by ray refinement.

Every surface and effective-node exchange conserves heat to below 10^{-10} W. Mapping surface heat back to package/reservoir nodes uses the same nonnegative temperature weights and preserves total power. Heat exchanged with the hot ceramic is included in the sink equation, not misapplied to the detector. Warm reservoirs are held at ambient; cap temperature gradients and their external cooling are not solved.

Independent two-plate gray-radiation, equal-temperature, zero-emissivity, vacuum, low-pressure scaling, high-pressure network, geometry-ownership and blocked-ray controls check implementation. Rejected material-domain and temperature-bound cases remain explicit. Exact reciprocal closure is a necessary consistency property, not proof of a physically accurate cavity operator.

14 Operating scenarios, electrical demand and rejected states

All reduced scenario families use 300 K ambient and 2 mW die heat. The low-parasitic family has vacuum, cap/window emissivities 0.08/0.05, 0.5 $\mathrm{K\,W^{-1}}$ sink and 0.75 times nominal Au conductance. The nominal family has 30 kPa Kr, emissivities 0.2/0.1 and a 2 $\mathrm{K\,W^{-1}}$ sink. The high-parasitic family uses 101.325 kPa N_2, emissivities 0.65/0.6, a 5 $\mathrm{K\,W^{-1}}$ sink, 1.25 times Au conductance and 1.5 times gas-network conductance. Several hypotheses change together; these families are not probability bounds.

At 0.15 A	Die (K)	Terminal (V)	Input (mW)	Sink (mW)
Low-parasitic vacuum	230.975	1.15443	173.164	179.253
Nominal Kr	232.209	1.15193	172.790	182.196
High-parasitic nitrogen	235.031	1.14554	171.830	187.787

These values use nonlinear 32-interval legs and the 65,536-ray filled-core exchange proxy. They are conditional engineering predictions. At the nominal point the hot ceramic is 300.364 K, intermediate temperatures are 268.059 and 246.537 K, and the cold ceramic is 232.104 K. The die is warmer by approximately 0.105 K because its attach carries both imposed and parasitic heat.

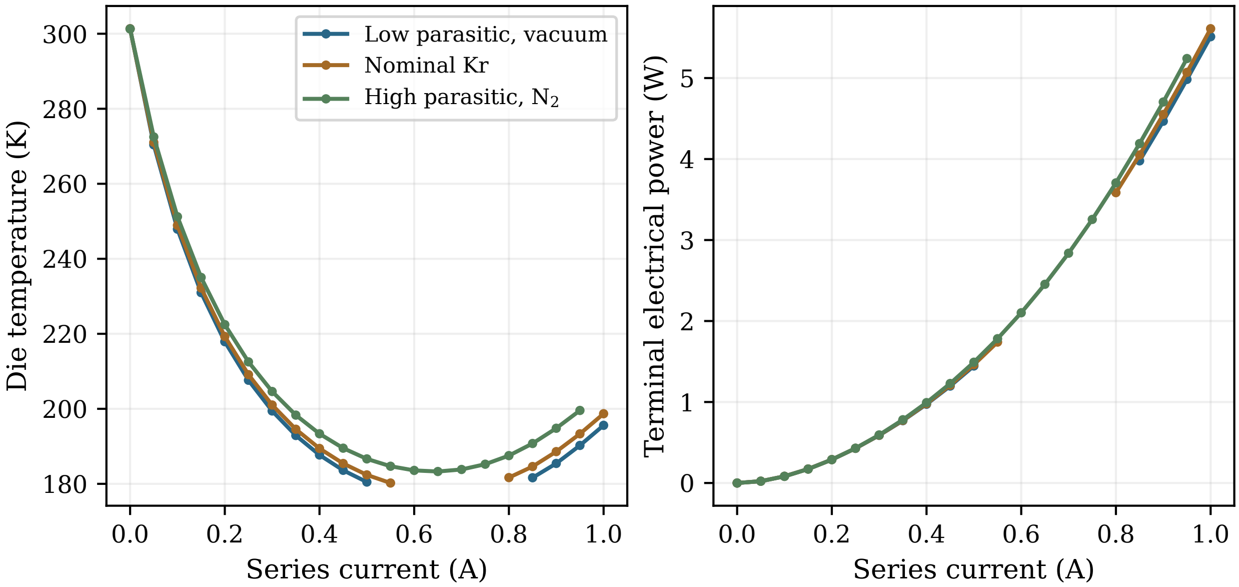

Figure 18: Reduced nonlinear/proxy scenario sweeps from 0 to 1 A in 0.05 A increments. Missing intervals are numerical rejections, not interpolated operating data. Curves do not represent three-dimensional cavity solutions, measured performance or qualified continuous optima.

Of 63 attempted points, 52 pass the reduced-model numerical gates and 11 remain rejected. Ten are lower-bound residual failures; the high-parasitic 1 A attempt rejects a leg profile at its active temperature domain. A failed candidate does not prove that no physical operating state exists. The imposed 180 K model floor also prevents interpreting accepted grid minima as unrestricted cooling optima. No source curve is silently extended below that floor.

14.1 Material and closure comparisons at matched current

At nominal 0.15 A, the constant P/N proxy with the older effective-reservoir model gives die 243.280 K and 0.84437 V. The source analog with affine temperature gives 231.202 K and 1.15539 V. Direct nonlinear transport with the same older reservoir closure gives 231.192 K and 1.15642 V. Adding the current filled-core closure gives 232.209 K and 1.15193 V. These alternatives change grade assumptions and/or package reduction; none establishes which hypothesis matches an unknown manufactured unit.

The approximately 12 K constant-to-analog difference is far larger than the 0.010 K affine-to-nonlinear change at this low-current point. At higher current the affine approximation can become materially inaccurate. The enclosure change at 0.15 A raises the nominal die by approximately 1.018 K; that shift includes changed surfaces, chords, reservoir paths and visibility assumptions, rather than a single independently measured physical mechanism.

The constant solid-only reduced case at 0.1 A and 2 mW gives 257.756 K and 0.58561 V when the hot ceramic is held at 300 K. The genuine three-dimensional case holds only the Kovar underside at 300 K and gives 265.593 K and 0.55194 V on its original mesh. Their difference is not a matched-boundary error estimate: spatial spreading, bypass and joint/header paths differ. It is retained as a warning about reduced-model structural omissions, not fitted away.

14.2 Material screening at a fixed hot ceramic

A separate preregistered screening keeps the hot ceramic at 300 K and die heat at 2 mW. External Cu/Au links are either present or entirely removed, including their electrical resistance and Joule heat. Gas, radiation and nonlocal Cu/alumina leakage are OFF; the warm wire ends share the hot-side reservoir. This is distinct from the 0.15 A, finite-sink krypton proxy scenario above.

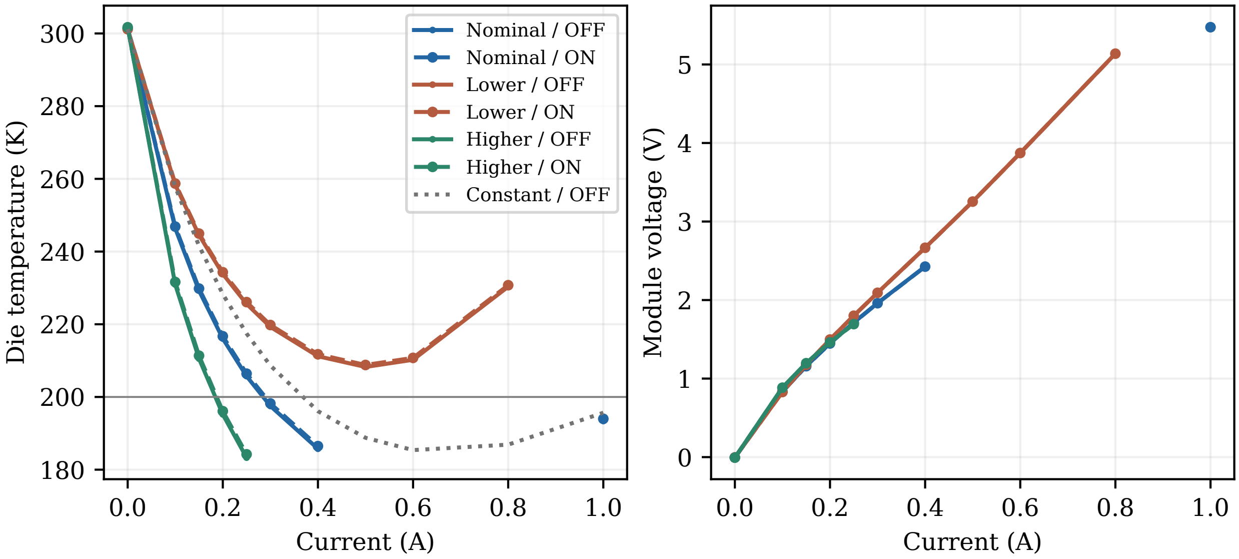

Figure 19: Frozen admitted samples for the three material hypotheses and external wires ON/OFF. The horizontal line marks 200 K; gaps retain rejected cases. The right panel shows module voltage, excluding external lead loss. The constant-property control is a separate baseline. Curves are reduced estimates, not measured performance or qualified continuous optima.

Sample	I (A)	T_d (K)	Terminal (V)	Input (W)	$Q_{c,cold}$ (mW)
Nominal	0.4	186.484	2.426999	0.970800	3.871307
Lower	0.5	208.804	3.255623	1.627812	3.503367
Higher	0.25	184.229	1.694506	0.423627	3.908485

These are the lowest admitted *sampled* wire-ON die temperatures, not continuous optima. Nominal and higher samples stay inside measured leg coverage; the lower sample reaches a 302.867 K internal peak and uses assumed continuation. Nominal $Q_{c,cold} = 3.871307$ mW includes Au heat as well as the imposed 2 mW. Thus die COP, $Q_{die}/P_{terminal} = 0.002060$, differs from cooler COP, $Q_{c,cold}/P_{module} \simeq 0.003990$.

The nominal/higher hypotheses demonstrate a conditional $T_d \leq 200$ K target on this grid; the lower hypothesis does not. Actual reference-device feasibility remains unresolved because the grade, contacts and omitted heat paths are not identified. No previous different-law bypass shift is added to these predictions.

14.3 Admitted domain and rejected operating states

All 181 preregistered cases were attempted: 141 passed the unchanged reduced numerical gates and 40 were rejected. Of 130 admitted nonlinear cases, 56 stay entirely within measured material coverage and 74 require assumed continuation; the other 11 are constant-property controls. No registered point was omitted for budget reasons.

Family	Accepted	Rejected
Material and wire grid	46	20
Constant-property controls	11	0
Hot-side/load controls	20	4
One-factor endpoints	12	0
Literal manufacturer conditions	46	14
300 K reference controls	4	2
Attach invariance	2	0
Total	141	40

Seven rejections fail the leg residual/domain check and 33 fail package residual/bound acceptance. Last available temperatures, residuals, active bounds and optimizer diagnostics are retained; they are not reported as operating observations. In the nominal grid, 0.5/0.6/0.8 A points reject while 1 A can be admitted on a warmer branch. Disconnected accepted samples therefore cannot establish a global optimum, a continuous branch or physical infeasibility. The active 180 K floor and nonlinear initialization constrain admitted coverage. Neither Q_{max} nor ΔT_{max} was computed.

Six independent same-grid bounded boundary-value checks, covering all three triples and two measured-domain stage conditions, differ by at most $2.18\,10^{-13}$ W in Q_c and $1.01\,10^{-13}$ V. They test implementation agreement, not experimental accuracy or continuum discretization error. Across admitted package cases the maximum global energy residual is $5.32\,10^{-11}$ W and stage error $2.90\,10^{-12}$ W, within the unchanged 10^{-8} W package and 10^{-9} W stage gates. Such conservation cannot qualify spatial fluxes or compensate for omitted device physics.

14.4 Manufacturer plausibility under explicitly matched boundaries

The 2019 TEC Microsystems 3MC04-044-05 vacuum charts [2] specify nominal 27 °C, interpreted literally as 300.15 K. The conditional panel matches stated current and prescribed cold load and compares *cold-ceramic lift* and *module voltage*; external wires and side exchange are OFF. It does not compare a loaded detector temperature with module-only zero-load lift.

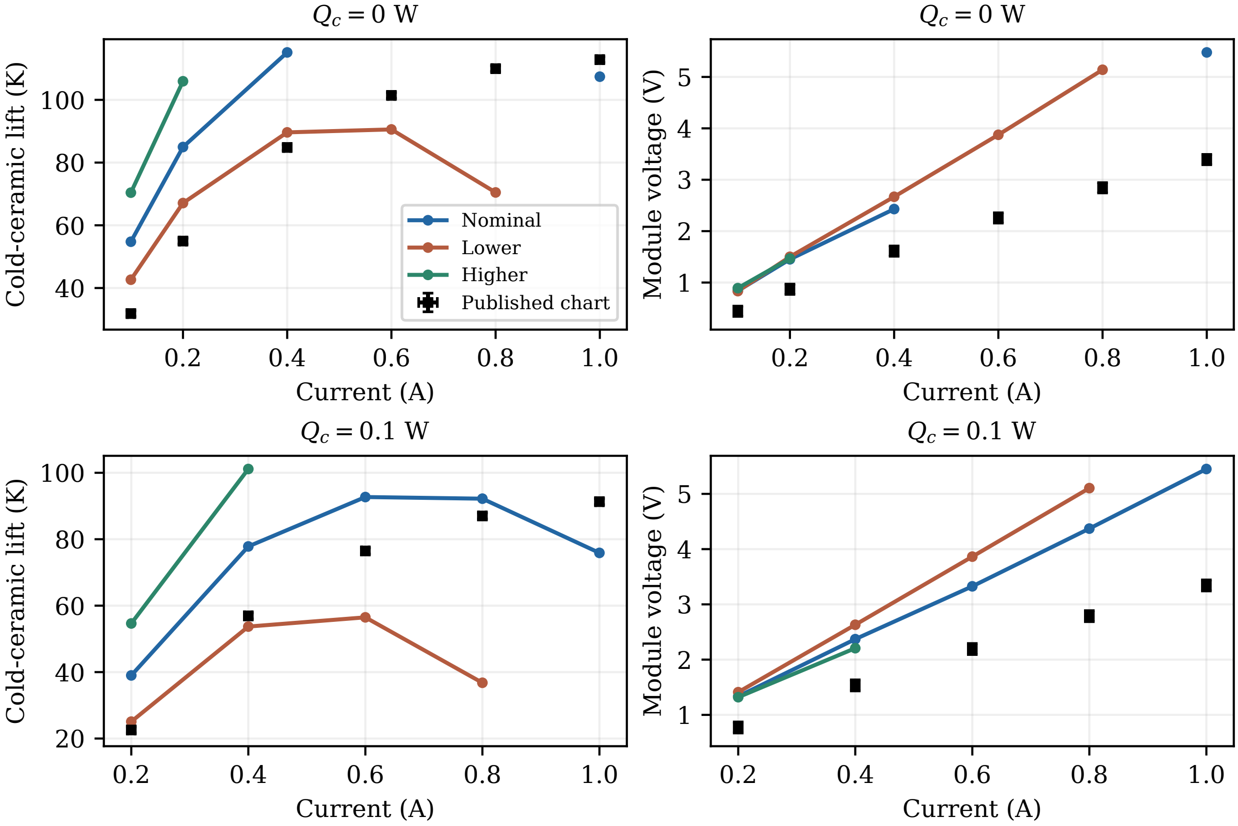

Figure 20: Conditional idealized vacuum module comparison at literal 300.15 K. Symbols are existing published chart readings with digitization bounds, not manufacturer uncertainty bars. Lines are frozen reduced calculations; rejected samples remain gaps. All model points require assumed above-300 K continuation. The discrepancy is not empirical agreement or commercial-grade identification.

Zero load, 0.2 A	Model	Chart	Difference	Relative (%)
Cold-ceramic lift (K)	84.958002	54.967287	+29.990715	+54.56
Module voltage (V)	1.452347	0.869116	+0.583231	+67.11

Across the ten common admitted current/load conditions, nominal mean absolute errors are 17.135582 K in lift and 1.058845 V. The 300-to-300.15 K control changes the 0.2 A zero-load result by only 0.074942 K and 1.294 mV; it cannot explain the approximately 29.991 K / 0.583231 V mismatch.

14.5 Limits of the manufacturer comparison

A strict measured-only comparison at 300.15 K is **not comparable**: the source tables end at 300 K. The panel is conditional on the engineering continuation, with local Joule overshoot also checked. A separate 300 K control is not an exact literal-27 °C match. Dry-N_2 charts are not comparable to this exchange-OFF calculation. Likewise, the 2 mW package-detector result is not a module-only zero-load measurement. Rejected cases have no accepted prediction; they are neither filled in nor hidden when assessing coverage.

Actual P/N grade, internal geometry and 28/12/4 allocation, contacts, vacuum pressure and radiative parasitics remain unidentified. Matching documented boundary variables and the observable does not establish equivalence of the physical module. Prescribed-load attach controls change die temperature while cold ceramic and module voltage remain invariant within 10^{-10} K and $3\,10^{-13}$ V, respectively; they do not validate real contact properties.

All chart subsets have already been examined; retained calibration/holdout labels do not create a fresh blind validation. Earlier identification found rank eight of nine for a different model, illustrating non-uniqueness rather than determining a grade. Its historical lift/voltage residual is not the current discrepancy. Simultaneous changes in grade, construction and missing paths cannot be attributed uniquely from this panel. The scaling analysis below proves a geometry/material ambiguity; independent construction and contact evidence remains necessary before interpreting a favorable grade adjustment.

14.6 Meaning of apparent numerical precision

Several tables retain extra digits to distinguish solved states and audit balances. Without identified material grades, contact impedances, die loading, accommodation and reservoir conditions, such digits do not provide a physical accuracy interval. A microkelvin-scale change in an output can coexist with a failed raw spatial flux criterion, as the solid refinements demonstrate. A small reduced-model residual similarly cannot certify a real detector temperature.

14.7 Geometry hypothesis: residual direction and physical realizability

At literal 300.15 K, zero cold load, external wires and exchange OFF, the common geometry factor changes both lift and voltage coherently. It is exactly transport-equivalent to reciprocal bulk factors $(r, h) = (1/g, g)$ with unchanged S. The manufacturer comparison therefore cannot identify which interpretation is physical. These fixed hypotheses were not optimized against the readings.

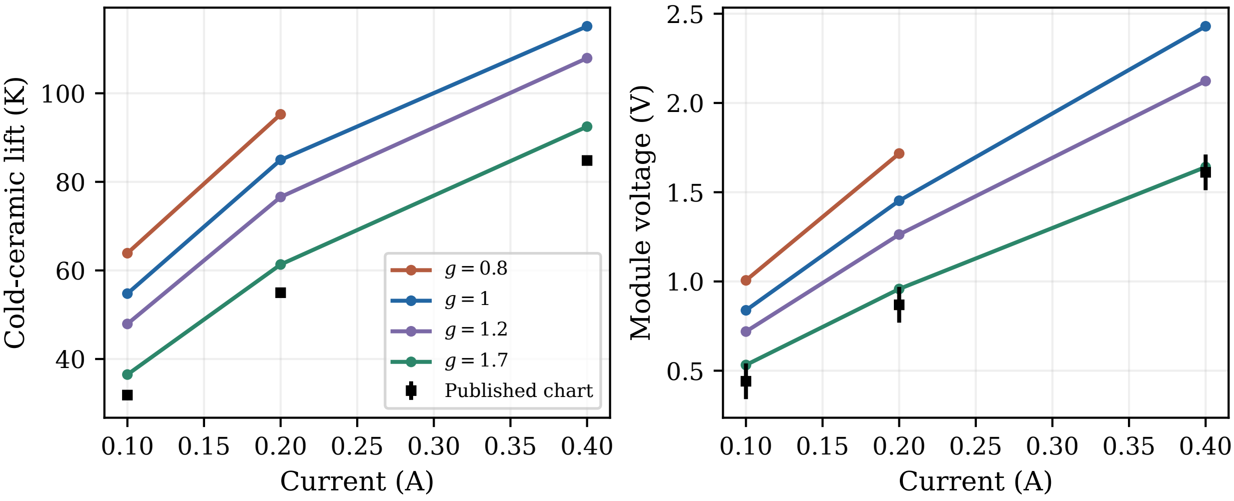

Figure 21: Admitted zero-load module samples at literal 300.15 K; gaps retain rejected points. Symbols show published chart values with vertical digitization bounds. All model points use assumed above-300 K continuation. These are conditional reduced estimates, not observations of changed native geometry or measured commercial grades.

At 0.2 A: g	Lift (K)	Module V	Lift difference (K)	V difference
0.8	95.247199	1.716908	+40.279912	+0.847792
1	84.958002	1.452347	+29.990715	+0.583231
1.2	76.579707	1.262872	+21.612420	+0.393756
1.7	61.339405	0.957962	+6.372118	+0.088846

Differences are model minus chart: positive means *overcooling and overvoltage*, not agreement. At $g = 1.7$, residuals remain approximately +6.372 K and +0.088846 V. Four common admitted source conditions give nominal versus $g = 1.7$ lift MAE 22.555 versus 7.393 K and voltage MAE 0.5951 versus 0.08065 V. One registered 0.1 A/0.1 W source reading is absent and is not interpolated. Higher-temperature dry-N_2 charts remain not comparable. Published 2.9-ohm AC resistance is not DC thermoelectric V/I.

The physical realization separately fails on unchanged native interfaces. Area ×1.7 increases width from 0.400 to 0.521536 mm; the minimum gap is 28.464 μm, but all 176 axial faces lose full Cu footprint coverage, with minimum 58.8235%. Shortening all legs by 1/1.7 instead breaks axial mating and gives a repacked height 2.758235 mm if other layers stay fixed, outside the candidate 3.8±0.15 mm envelope. This neither qualifies altered CAD nor proves every alternative manufactured design impossible.

In the separate 300 K, 2 mW die, wires-ON control, $g = 1.7$ admits 198.492 K at 0.5 A and 0.967238 W input. It is not an exact 200 K point or power optimum. External warm series resistance raises voltage; a cold-face thermal resistance changes loaded-face temperature but has no zero-load effect in that limiting placement. Unknown internal contacts, finite-sink matched-target power and independently measured dimensions remain future questions; no fitted grade or actual package feasibility is established.

15 Heat budgets and dominant paths

At nominal 0.15 A, total die heat is 3.507 mW: the 2 mW imposed source plus approximately 1.507 mW gas/radiation/Au input. The cold ceramic also absorbs approximately 2.614 mW, so the coldest stage removes approximately 6.121 mW. Interstage loads accumulate toward the hot stage.

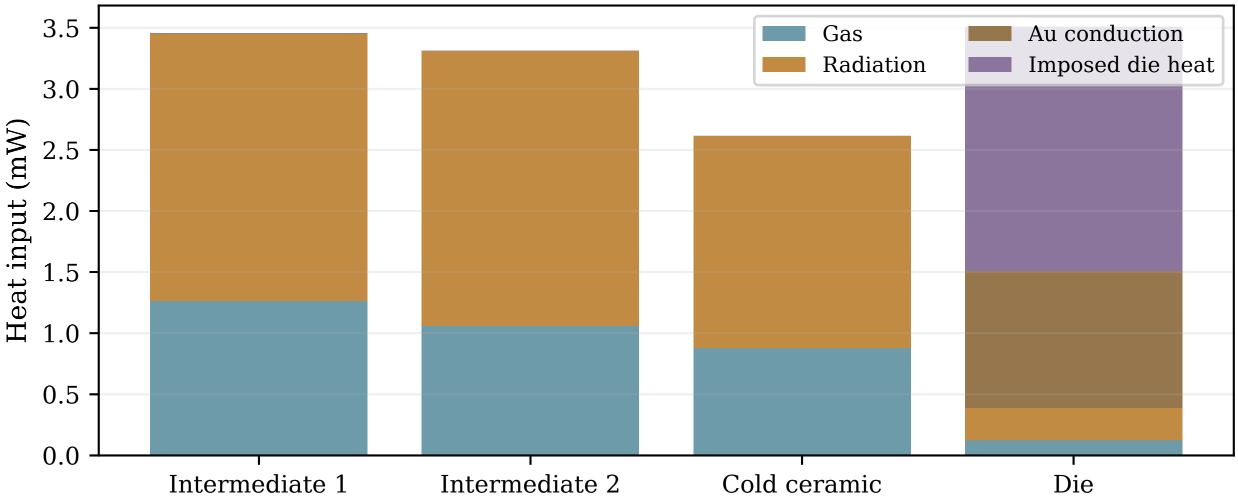

Figure 22: Net positive heat loads at the four cold/intermediate reduced nodes for nominal 0.15 A Kr. Attach transfer and interstage thermoelectric flows are internal and are not stacked again as external loads. Vertical proxy surfaces distribute heat using their stated node weights.

Stage	Q_c (mW)	Q_h (mW)	V (V)	Joule heat (mW)
1	56.508	185.594	0.86057	76.029
2	17.432	53.053	0.23748	21.241
3	6.121	14.119	0.05332	4.921

Whole reduced device, signed inputs	Power (mW)
Terminal electrical input	+172.789624
Imposed die heat	+2.000000
Net header/cap/window heat into device	+9.021852
Au warm-end conduction into die	+1.117526
Cu pin-end thermal input (signed)	-2.732546
Heat rejected at effective sink	+182.196456

The first five signed entries sum to sink rejection. Warm-reservoir power is explicitly accounted for; it is not part of terminal electrical input. The hot-side Cu connection carries heat outward because the pad is warmer than its ambient pin. The nominal global discrepancy is below 10^{-11} W; the maximum over accepted proxy grid points is approximately $1.41\,10^{-11}$ W. Those figures establish bookkeeping consistency, not a physical heat-leak uncertainty.

16 Sensitivity, model-form uncertainty and physical interpretation

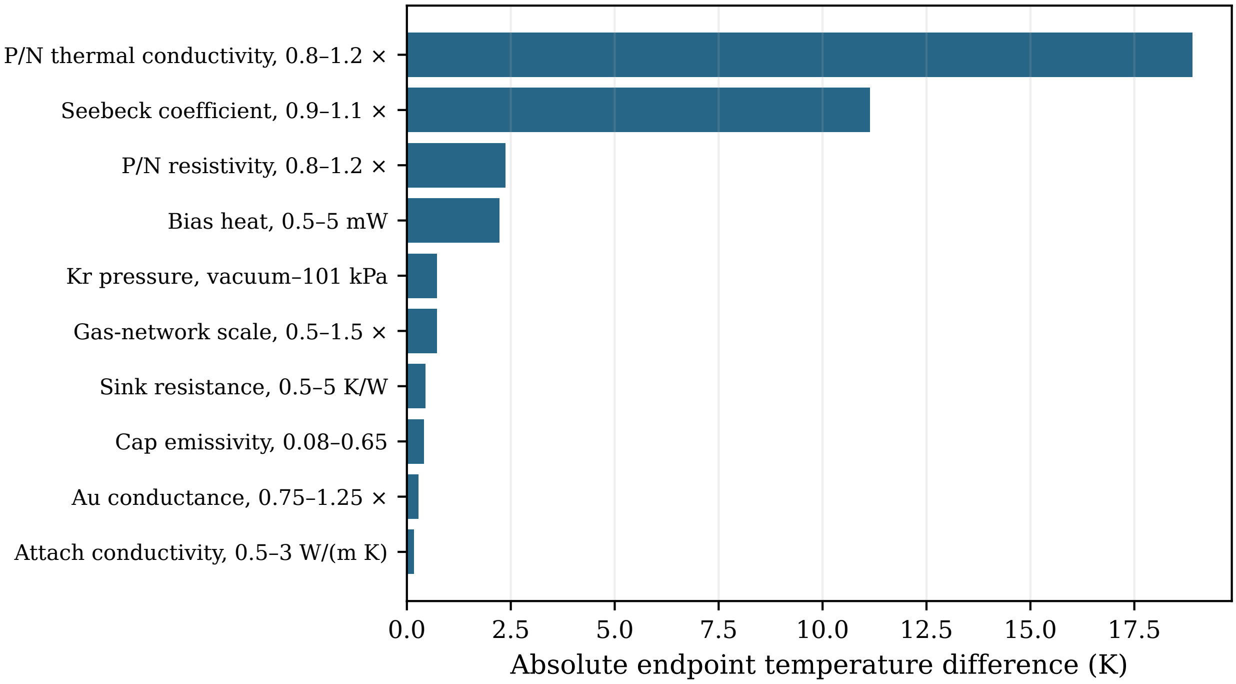

Figure 23: One-factor engineering endpoint spans about nominal 0.15 A, with nonlinear transport and the filled-core closure. Trial intervals are shown explicitly. They are chosen hypothesis ranges, not probability distributions or confidence intervals. Different trial widths would change the ranking.

P/N thermal conductivity scaled from 0.8 to 1.2 times the selected law changes the die by 18.900 K; Seebeck scaled from 0.9 to 1.1 changes it by 11.142 K. Resistivity scaled 0.8–1.2 gives 2.371 K, and bias heat from 0.5 to 5 mW gives 2.226 K. In this particular closure, vacuum to 101.325 kPa Kr spans 0.721 K, sink resistance 0.5–5 $\mathrm{K\,W^{-1}}$ spans 0.446 K, and cap emissivity 0.08–0.65 spans 0.413 K.

These results support prioritizing grade/contact and loading hypotheses when interpreting absolute cooling. They do not prove that real gas or radiation is negligible: the proxy omits interleg spaces, pin/wire occlusion, nonisothermal cap behavior and physical gas diffusion. Its geometry/model-form error is distinct from the reported ray-refinement changes. Ideal contacts and zero additional stage bypass also leave unresolved paths that can change both cooling and voltage.

Unknown construction and operating inputs should therefore be treated as alternative engineering hypotheses. Independent thermal/electrical measurements would constrain them, but absent proprietary data need not prevent bounded synthetic studies. Empirical comparison, when available, requires matched boundary temperature, current, imposed load and ambient conditions. No inferred confidence interval or commercial performance claim follows from the present sensitivity spans.

16.1 Controlled input spans in the independent material screening

The exchange-OFF, wire-OFF nominal control at 0.25 A, fixed hot ceramic 300 K and 2 mW die heat gives $T_d = 205.722029\,\mathrm{K}$. Each row varies only its stated input; the other laws, dimensions and conditions remain fixed.

Input interval	Absolute endpoint T_d span (K)
k scale 0.8–1.2	23.348849
S scale 0.9–1.1	14.172527
Hot ceramic 290–310 K	9.084846
ρ scale 0.8–1.2	4.959973
Die heat 0–5 mW	2.219005
Attach k 0.5–3 W/(m K)	0.100000

Turning the wires ON at the same control adds 0.684866 K. These are mechanistic paired contrasts, not probability ranges or a confidence ranking. The widths and units differ: transport spans per percentage point are approximately 0.709 K for S, 0.584 K for k and 0.124 K for ρ. These secants are not local derivatives, and their ordering differs from the raw endpoint spans. Neither comparison measures unknown manufacturing scatter.

The strong conductivity and thermopower effects motivate independent material and construction evidence. They do not establish which physical grade should be selected, whether the assumed joint property changes are manufacturable, or whether actual gas/radiation and nonlocal leakage are small. The earlier 0.15 A proxy sensitivity uses different boundaries and remains a separate study; its endpoint spans are not silently replaced by these controls.

17 Overall verification and engineering conclusions

Assessment	Result	Interpretation
Three-dimensional geometry/current	PASS numerical checks	Corrected mesh and 177 interior current cuts
Three-dimensional weak heat/free residuals	PASS numerical checks	Algebraic and control-volume consistency
Original cold/bath raw criteria	**FAIL** overall	Initial 6.48% cold; best normal bath 1.069%; later transverse about 1.29%
Same-mesh release/comparator	Not demonstrated	Both raw limits and all other checks are still required
Bounded nonlinear leg transport	PASS tested comparisons	Same-grid/analytical/refined numerical checks; not grade validation
Reduced package conservation	PASS numerical checks	Accepted states meet unchanged energy/domain gates; rejected candidates retained
Proxy visibility/reciprocity/closure	PASS consistency checks	Closed filled-core rays and balanced exchange; 0.04043 maximum last-level correction
Native full-cavity exchange	Not implemented	No integrated gas diffusion, all-surface view factors or coupled cavity field
Material screening	141 accepted / 40 rejected	56 measured-domain nonlinear; 74 continuation; 11 constant controls
Standalone gas raw flux	**FAIL**	32.6274% mismatch against 1%; not a qualified device heat budget
Material and device accuracy	Not established	Large joint manufacturer discrepancy; grade/construction uncertain

17.1 Engineering conclusions

The corrected 88-leg solid calculation passes electrical and conservative heat checks while retaining failed raw bath/cold criteria. Stable temperatures and voltages do not establish numerical release. Bounded nonlinear transport improves high-current agreement with an independent one-dimensional reference; neither that agreement nor algebraic conservation validates a physical material grade.

Two reduced operating contexts must remain distinct. The 0.15 A nominal krypton proxy gives approximately 232.2 K, 0.173 W electrical input and 0.182 W sink rejection. The independent fixed-hot, exchange-OFF screening admits nominal 186.484 K at 0.4 A, wires ON and 0.970800 W input, wholly inside measured leg coverage. Its 200 K target is demonstrated only under those admitted hypotheses, not for an actual assembly. Large joint manufacturer lift/voltage discrepancies, unknown contacts and omitted nonlocal paths prevent a commercial or physical-accuracy claim. No continuous optimum or device confidence interval is established.

The filled-core proxy conserves power while omitting native transport. The separate gas control retains its 32.6274% raw failure against 1%; those powers are not coupled here. Nonlinear scaling ambiguity and failed unchanged-pad realization preclude an inverse dimension claim. Independent geometry/material/contact evidence is required; no new coupled field or numerical release follows.

17.2 Reproducible scientific method

Three-dimensional figures are exact intersections of archived tetrahedra and solved fields, with independent node/cell correspondence and voids preserved. Reduced plots use completed cases and independent boundary-value comparisons; rejections remain visible. Candidate geometry and proxy are audited separately. Source/input identities, field provenance and deterministic figure records are retained independently. Rebuilding figures implies no new physics solve.

References

[1] VIGO Photonics, *3TE-TO66(9p)-wW PVI/PCI Detector Technical Drawing*, ZTM-TO66-Z030, Revision 9, 1 March 2024. Public family drawing; no identified composite assembly.

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[3] E. M. Heckman, L. P. Gonzalez and S. Guha, *Measurement of optical and thermal properties of* $Hg_{1-x}Cd_xTe$, Applied Optics **47**, 578–582 (2008), doi:10.1364/AO.47.000578. Composition-dependent thermal model.

[4] J. Zhang et al., *Accurate Measurements and Error Analysis of* Bi_2Te_3*-Based Low-Temperature Thermoelectrics*, *Chinese Physics Letters* **42**, 056304 (2025), doi:10.1088/0256-307X/42/5/056304. Digitized P/N specimen transport, 40–300 K; no commercial grade identification.

[5] M. L. Huber, *Thermal Conductivity of Gases*, NIST/CRC (2019), NIST thermophysical-property tabulation. Dilute-gas conductivity data; package transport is separately approximated.